

# Context selection as a partial order: A Blackwell framework and verified LLM evidence

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## Abstract

Choosing what to put in a language model’s prompt is almost always done by scoring each candidate source with a single number and keeping the top few. We argue this is the wrong approach. The best context for one task is often useless for another. Two sources can be *incomparable*, in the exact sense of Blackwell’s 1953 comparison of experiments, and when they are, no single per-source score, relevance or utility, can rank them correctly for every task. To test this on real systems we use a simple check: run each source through a battery of pass/fail tasks, compare the pass rates, and attach confidence intervals so the comparison is auditable (a distribution-free *verification*). On six language models from four vendors we confirm ( $\geq 90\%$  confidence, both directions) that two hand-built sources are incomparable. On a designed trap task a lexical score, two dense retrievers, and a cross-encoder reranker all prefer the wrong source; only a listwise LLM reranker, which reads the deciding detail, gets it right, exactly the boundary the theory predicts. We also find that adding a strictly larger source can *hurt*: pass rates fall from 100% to as low as 0%, and controls rule out word order and length as the cause. The lesson is practical: treat context selection as a partial-order problem, not a scalar ranking, and compare sources by a verified pass-rate test rather than a relevance number. Scaling the selection rule this implies to large corpora is the main open problem.

**Keywords:** context selection; Blackwell order; comparison of experiments; incomparability; partial order; retrieval-augmented generation; large language models; distribution-free verification; anti-monotonicity; prompt engineering

## 1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. This everyday situation is the entire paper in miniature. It has a precise name, the two notes are *incomparable* in the sense of Blackwell’s comparison of experiments [Blackwell, 1953], and a precise consequence: no single relevance score can rank the two notes so that the ranking is right for every task, because a score must crown one winner and here neither note wins outright. The body makes this exact, gives a statistical test that *verifies* it from ordinary pass/fail runs on a live model, and finds that six production language models across four vendors all exhibit it. A

reader who keeps these two notes in mind can read every symbol below as a statement about the first note versus the second (in the body they are the sources  $W_1$  and  $W_2$ ).

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information,  $\mathcal{V}$ -usable information [Ethayarajh et al., 2022], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. We argue it is also the wrong abstraction, and that a classical apparatus from statistical decision theory says precisely why.

**A folklore observation, restated as structure.** Practitioners already know the phenomenon: *relevance does not transfer across tasks*. A source that helps one task hurts another, so “the best context” depends on the task, and this is usually treated as a nuisance to engineer away with better rankers. We show it is instead the signature of a *partial order*. Model a context source  $W$  as a statistical experiment on what the task actually needs,  $S$  (the thing a verifier checks; §3). Blackwell compares two experiments by asking whether one is a blurred copy of the other: we write  $W_1 \succeq_B W_2$  when  $W_2$ ’s signal is a noised version of  $W_1$ ’s, and by the Blackwell–Sherman–Stein theorem this holds exactly when an ideal decision-maker does at least as well with  $W_1$  as with  $W_2$  on *every* task [Blackwell, 1953, Torgersen, 1991]. The catch is that the order is only *partial*: often neither source is a blurred copy of the other, so the two are *incomparable*. A single score cannot capture a partial order. A score forces every source into one line, but two incomparable sources each win on some task, so whichever way the score orders them, it is wrong on one.

A note on the scope of the title. A total order is refuted by a single incomparable pair, so the claim in the title is licensed by exhibiting verified incomparable pairs, which we do on six production models across two source domains (§7). Beyond that existential core, the partial order is offered as a *framework* for the general phenomenon, and every empirical claim in the paper is stated model-relative, not as a law over all context selection.

**What is and is not new.** The mathematics is classical: the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam’s deficiency. Several of the *phenomena* are also known, each in its own literature, and we say so up front. That topical relevance is the wrong target and downstream utility the right one is the thesis of utility-centric retrieval [Zhang et al., 2026a]; that the same passage is worth different amounts to different models is measured there [Zhang et al., 2025]; that adding sources can lower accuracy while coverage saturates was established for data integration over a decade ago [Dong et al., 2012]; that rerankers are steered by lexical similarity is documented in the wild [Hagström et al., 2025]; and keeping a non-dominated set rather than a top- $k$  prefix exists as an operator over heuristic scores [Zhu et al., 2026]; and the Blackwell and Le Cam apparatus itself was brought to inference-time context selection against a frozen decoder, concurrently with this work and without a priority claim on our part [Alpila, 2026a]. What is new is the *form* that ties these together: reading cross-task non-transfer as a partial order over context sources; proving that no single per-source score, relevance or utility, can be faithful to that order across tasks once two sources are incomparable, so that utility re-measured per task by running the model is not a shortcut around the theorem but the probe it calls for; and giving a finite-sample test that verifies incomparability and superset harm on a live model with both sources held correct and the length, position, and routing confounds controlled (§2). One gap needs care. Blackwell’s theorem is about the *best possible* use of a source, but every number we measure comes from one

fixed model:  $\text{PASS}_M(W, T) = \mathbb{E}[v_T(\delta_M(T, w))]$ , the pass rate of a single rule  $\delta_M$  that is not optimal. So we state the order twice, once for the ideal user and once for the fixed model, and prove a one-way bridge from what we measure to what the theorem is about (§4); and we call our estimator a *value-deficiency surrogate*, not a Le Cam deficiency (§6).

**A latent-coordinate model that makes the order non-vacuous.** One technical point keeps the framing from being empty. A fixed string of text looks like a constant, and a constant carries no information in Blackwell’s sense. We avoid this by treating the hidden requirement as a list of separate facts and each source as revealing some of them; two sources are then incomparable exactly when each reveals a fact the other hides, which is easy to check and holds for our pair (Lemma 1).

**Empirical findings.** We test the theory on the two notes from the opening example: an API call contract  $W_1$  and a wire-encoding rule  $W_2$  for the same module (§7), graded by a small battery of tasks whose verifier returns pass or fail (exit code 0 is pass). The battery is a fixed, hand-built opposition, not a random sample of tasks, and the verified result rests on one API task and one encoding task; we say so plainly and do not call it a “distribution.” On this setup, from *real* model calls with no invented numbers, we find:

- **Verified empirical incomparability on all six models.** The uniform verification clears  $\geq 90\%$  confidence in *both* directions on every model, with two-sided lower bounds from +0.35 to +0.76 at  $n=40$  (§7.4.1), so incomparability is not a single-lab artifact.
- **Representative topical rankers mis-rank the pair.** On a trap task saturated with encoding vocabulary, lexical relevance ranks  $W_2 \succ W_1$  while realized PASS ranks  $W_1 \gg W_2$  (90–100% vs. 0%); two dense bi-encoders and a cross-encoder repeat the error, and an LLM listwise reranker escapes it (§7.4.2). The lexical and dense rankers agree with PASS on both honest cells, so their failure occurs *only* where surface relevance diverges from verifier-decisive content, which is the boundary Proposition 2 draws.
- **Verified anti-monotonicity.** The superset  $W_{1+} = W_2 ++ W_1$  (the operator  $++$  is text concatenation) literally *contains*  $W_1$ ’s signal yet sharply degrades the API cells: the harm contrast is verified on at least one cell for all six models, as deep as  $100\% \rightarrow 0\%$  (upper bound  $-0.82$ ) and as mild as Opus’s  $100\% \rightarrow 38\%$  ( $-0.37$ ); a length-matched padding control shows equal-length irrelevant text is harmless, so the harm is the added signal, not token count or position (§7.4.3). The effect is robust across the capability ladder but not monotone in capability, so we claim no capability trend.
- **A supporting observation: private-context value is model-relative.** The none-arm baselines span 0%–75% guessability across vendors (§7.4.4), an empirical instance of the theory’s  $\theta$ -dependence, reported as supporting evidence rather than a headline claim.

**Contributions.** The paper makes one claim, that the order over context sources is *partial* rather than total, and contributes the three pieces that claim needs: what follows from it, the instrument that makes it testable, and the evidence that it holds.

- (C1) **Theory: the order is partial, and no single per-source score survives it** (§4, §5). We formalize context sources as statistical experiments on the task’s latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the  $\delta_M$ -restricted order the experiments probe, and

prove a one-way bridging lemma between them. Bringing this apparatus to inference-time context is not by itself ours: concurrent work reaches the same classical objects from the admission side (§2.3). What is ours is the consequence of taking *incomparability* rather than deficiency-minimization as the object. Blackwell-incomparable sources cannot be ranked by *any* single per-source score  $\varphi(W)$ , relevance or utility, for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2); both impossibility results are machine-checked in Lean 4 and depend on no axioms (Remark 3). Information retrieval has long argued that no single fixed document score is optimal once documents interact (§2.1); we derive the impossibility as a theorem in the Blackwell frame and name the exact structural condition under which it bites.

- (C2) **Verification: a measurable deficiency surrogate with finite-sample guarantees** (§6). We define  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)])$ , a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform verification: exact Clopper–Pearson intervals [Clopper and Pearson, 1934] at Bonferroni level [Bonferroni, 1936]  $\alpha = \eta/(2|\mathcal{D}|)$ , valid jointly on a single event, so that  $L_{\mathcal{D}} > 0$  verifies a positive value gap at confidence  $\geq 1 - \eta$ ; we argue why a naive DKW bound [Dvoretzky et al., 1956] under-covers the supremum, and give the matching interference verification. This is the instrument that makes the partial order testable on a real model behind a real verifier.
- (C3) **Evidence: verified incomparability and scalar-ranking failure on live models** (§7). Verified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and verified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in §8.4.

**Roadmap.** §2 positions the work; §3 fixes the objects and defines  $\theta$ ; §4 develops the partial order, the  $\delta_M$ -restricted order and the bridging lemma; §5 proves the impossibility and isolates the query-conditioned case (C1); §6 gives the surrogate and its verifications (C2); §7 reports the six-model program (C3); §8.4 states scope, and §9 concludes. The two impossibility proofs accompany their statements; the rest is deferred to Appendices A–C. Every reference has been verified against arXiv/DBLP/publisher records, and both impossibility results are machine-checked in Lean 4 (Remark 3).

**How to read the formal terms.** The argument is necessarily phrased in the language of statistical decision theory. Table 1 gives the engineering reading of each recurring term; the empirical sections (§7 onward) can be followed with the right-hand column alone.

## 2 Related Work

**Positioning.** The tools we use are classical, Blackwell’s garbling order [Blackwell, 1953] and Le Cam’s deficiency [Le Cam, 1964, 1986]; we add no new core mathematics, and bringing them to inference-time context is concurrent with, not prior to, Alpila [2026a] (§2.3). Our contribution is to take *incomparability* as the object and make it measurable.

| Formal term                          | Engineering reading                                                                                                                              |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| source $W$ (an experiment)           | a candidate context item: a doc, note, or module you could paste into the prompt                                                                 |
| Blackwell dominance $W \succeq_B W'$ | $W$ answers everything $W'$ answers; keeping $W$ and dropping $W'$ is safe                                                                       |
| incomparability                      | each source is decisive on tasks where the other fails; no single number can rank them once and for all                                          |
| garbling                             | $W'$ is a lossy transformation of $W$ : everything in $W'$ is derivable from $W$                                                                 |
| anti-monotonicity ( $\Delta_T < 0$ ) | pasting more true, on-topic text into a working prompt can break it                                                                              |
| non-dominated front                  | the shortlist you must keep whole, because nothing on it verifiably covers anything else on it                                                   |
| verification                         | a claim measured with exact finite-sample error bars, still significant after multiple-comparison correction, with no distributional assumptions |
| model rule $\delta_M$                | the model as it actually behaves on the prompt, not an idealized optimal user of the information                                                 |
| surrogate $\hat{\delta}_D$           | how many PASS points you lose by using $W'$ in place of $W$ on the task family                                                                   |

Table 1: Glossary: the recurring formal terms and their engineering reading.

## 2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top- $k$  prefix: topical relevance, lexical or learned [Yu et al., 2024];  $\mathcal{V}$ -usable information [Ethayarajh et al., 2022, Yuan et al., 2025]; expected information gain and Bayesian experimental design [Choudhury et al., 2025, Chan et al., 2025]; game-theoretic and influence-based valuation of demonstrations [Xie et al., 2024, Vinay et al., 2024] and of retrieved documents [Nematov et al., 2025]; perplexity-reduction utility [Dai et al., 2025]; and binary sufficient-context detection [Joren et al., 2024]. Each collapses the comparison of two sources to one real number and is therefore a scalar of the kind the paper rules out (Theorem 2 for query-independent scores, Proposition 2 for topical query-conditioned ones): not a competitor to out-perform but an instance of the excluded class (we prove Blackwell-monotonicity for the closest neighbour,  $\mathcal{V}$ -usable information, in Proposition 1, and conjecture it for the rest). Guarantees do not change the class: conformal context filtering [Chakraborty et al., 2025] calibrates a distribution-free threshold *on* a scalar score and so inherits the total order it calibrates, whereas what we verify is an order relation between sources; and scorers of equal ranking quality can retain different document sets [Frohmann et al., 2026], a symptom of the same under-determination. The machinery for identifying a non-dominated set from noisy multi-objective samples is in turn classical in the bandit literature: Auer et al. [2016] give confidence-bound algorithms that identify the Pareto front from stochastic samples, with a gap-based sample complexity of the kind our verification pays for directly. What is new here is not the statistical machinery but the object it is run on, (model, verifier, task-battery) triples, and its use as a verification of incomparability between context sources.

The retrieval community has itself moved from relevance to *utility*, the downstream gain a passage brings [Zhang et al., 2026a], and has found that utility is model-specific [Zhang et al., 2025] and can depend on which other passages are present [Jain and Garimella, 2025]. We regard this as the same diagnosis reached from the other side, and the present paper as its formal boundary. Utility re-measured per task by running the model is not a fixed per-source score; it is the probe

our estimator is built from, and its cost is the price we make explicit. Utility *predicted* by a learned scorer is a number again: fixed across tasks it is covered by Theorem 2; query-conditioned but topical, by Proposition 2; only a scorer that reads the decisive coordinate escapes, which is the listwise boundary. Three families sit at the boundary. Listwise LLM rerankers [Sun et al., 2023] and learned set selectors [Jiang et al., 2026, Fan et al., 2026] are query-conditioned *set* functions, outside the theorem’s scope, though they carry no dominance guarantee of any kind. Skyline selection [Zhu et al., 2026] keeps every non-dominated candidate instead of a top- $k$ , the operator our own selection rule uses, but takes dominance over heuristic per-item scores rather than over task value. Diversity-aware selection, from MMR [Carbonell and Goldstein, 1998] to DPPs [Ye et al., 2023] and submodular context selection [Ghulyani et al., 2026], optimizes set objectives over similarity geometry rather than over verified task value. Directed-information  $\gamma$ -covering [Huang, 2025] is the closest of these in form, because its covering relation is genuinely *asymmetric* and information-theoretic rather than geometric: chunk  $i$  covers chunk  $j$  when it predicts  $j$  to within  $\gamma$  bits. The relation it orders, however, is redundancy *between sources*, computed query-agnostically and offline, so it prunes what another chunk already predicts; it never exhibits two sources each decisive on a task the other fails, which is the relation we measure.

The limits of scalar relevance are one of the oldest themes in information retrieval, and we position inside that lineage rather than apart from it: the Probability Ranking Principle [Robertson, 1977] is optimal only under an independence assumption, its failure once documents interact is long documented [Fuhr, 2008, Carbonell and Goldstein, 1998], and document interdependence was given ranking principles of its own, portfolio theory [Wang and Zhu, 2009] and the quantum probability ranking principle, whose “interference” term names the dependence of one document’s relevance on another [Zuccon and Azzopardi, 2010]. The LLM-era version, relevance-maximizing retrieval degrading answer quality, is measured directly [LeVine and Varjavand, 2025], and the companion observation that relevance does not imply *sufficiency*, so that missing hops and unresolved conflicts cannot be detected by scoring passages independently, is argued at the set level [Qiu et al., 2026]. Our delta is the *form* of the statement: the failure becomes a theorem in the comparison-of-experiments frame, with the precise structural condition, Blackwell incomparability, under which every query-independent scalar must mis-rank, and a verified live instance on six production models.

## 2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [Shi et al., 2023a]; position matters [Liu et al., 2024]; high-scoring distractors hurt while random noise can help [Cuconasu et al., 2024]; retrieval curves are inverted-U in passage count [Jin et al., 2024]; document count hurts at fixed length [Levy et al., 2025]; added length alone hurts [Du et al., 2025]; and the effect persists across model generations [Hong et al., 2025] and agent-scaffold components [Liu, 2026]. The genus is older than language models: in data integration, adding sources raises coverage while integration quality first rises and then falls, so that “less is more”: low-quality sources deteriorate the fused result rather than bringing the expected gain, and the right move is to select a subset before integrating [Dong et al., 2012]; the lesson has since been carried explicitly to augmented language models [Halevy and Dwivedi-Yu, 2023]. We do not claim to discover this genus. Two differences locate our finding inside it. Where data integration traces the harm to low-quality sources and restores monotonicity by redesigning the fusion rule, our two sources are individually correct and sufficient, the harm is verified against length, position, and formatting controls to be convention routing, and the rule  $\delta_M$  is a fixed model that cannot be redesigned, so the response is selection rather than fusion. And where “interference” in the mechanistic RAG literature means distractor passages steering a model, a failure recently localized and corrected in a sparse

feature space [Giannetti et al., 2026], and inter-context *conflict* means contexts that contradict each other [Xu et al., 2024], our sources neither distract nor conflict: both are true and both are needed, and the superset still collapses. Our anti-monotonicity finding contributes the order-theoretic form of this: a strict *superset* of a sufficient source, on sources separately *verified* Blackwell-incomparable, collapses PASS under a per-task verification of the direct contrast  $\text{PASS}(W_{1+}, T) < \text{PASS}(W_1, T)$ : a reversal no monotone context-value scalar can carry. A representation-level account of when context helps, an additive error decomposition with alignment conditions [Wang et al., 2026], is complementary: even in its multi-context extension it characterizes *when* a context corrects the error, not an order relation between candidate sources. Pareto-incomparability of prompts along competing metrics *within* a single task [Zhao et al., 2025] is orthogonal to our object, incomparable *sources across tasks*.

### 2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning, and, concurrently with this work, inference-time context selection itself. Alpila [2026a] formalizes the admission problem against a *frozen decoder* as minimization of Le Cam deficiency, pairs it with the Vereshchagin–Vitányi conditional structure function for the representation problem, proves a capacity floor below which the deficiency cannot be driven, and validates a curve-aware allocator on seven decoders in four families; a plain-language companion states the framing [Alpila, 2026b] and the artifact is released [Alpila, 2026c]. That work and ours reach for the same classical apparatus in the same month, and we make no priority claim over it. The objects are nonetheless different, and the difference is what this paper is about. There, deficiency is a scalar risk gap *inside one declared decision problem*, with the garbling theorem explicitly set aside as inactive there; the partial order that work does invoke runs over *compression methods* and is asserted structurally rather than measured. Our object is the order *between two admitted sources across tasks*: we show that order is partial by *measuring* mutual non-domination, derive the consequence that no single per-source score can be faithful to it, and verify a strict superset of a sufficient source collapsing PASS with the length, position and routing confounds controlled. Incomparability in this sense cannot arise inside a single declared decision problem, so its absence there is not a gap but a different question. A second line is training-*dataset* valuation: Zheng et al. [2024] order datasets by Blackwell informativeness and show that in their Bayesian regime a *scalar*, the mutual information between the curated and the test data, determines that order, the opposite of our finding that no query-independent scalar can be faithful once the order is partial (Theorem 2); the difference is that their comparison is taken against the Bayes-optimal user of the data, ours against a fixed model rule on a task family. Nearby uses stop short of what we do: garbling to harden LLM evaluations [Bradley, 2024], statistical deficiency to order training *tasks* [Fosse et al., 2025], a deficiency-induced *directional* dominance in transfer learning (no incomparability, no distribution-free bound, outside language modeling) [Akdemir, 2025], comparison-of-experiments ideas in loss design [van Rooyen and Williamson, 2014], and a survey of Blackwell’s theorems for AI [Paxton, 2026]. The one use of the order for language models at inference time that we know of sits on the other side of the model: Zhang et al. [2026b] rank the information structures induced by multi-agent protocols (voting, debate) and prove a pooled-posterior upper bound, a dominance result over protocols with no incomparable pair and no verification. To our knowledge no prior work *measures* a Blackwell partial order over the context sources of a single model, treats verified *incomparability* rather than deficiency-minimization or dominance as the object of interest, derives the no-per-source-score consequence from it, or supplies a distribution-free finite-sample verification of an order relation between named sources on production models. Those are the pieces that are new here.

| Method family (representative)                                                 | Cheap?         | Interaction?   | Incomparability? |
|--------------------------------------------------------------------------------|----------------|----------------|------------------|
| Scalar dense / PRP retrieval [Robertson, 1977]                                 | ✓              | ×              | ×                |
| Late interaction, multi-vector [Khattab and Zaharia, 2020]                     | ✓              | ×              | ×                |
| V-usable information [Ethayarajh et al., 2022]                                 | ✓              | ×              | ×                |
| LM-supervised retriever [Shi et al., 2023b, Guu et al., 2020]                  | ✓ <sup>a</sup> | ×              | ×                |
| MMR diversity [Carbonell and Goldstein, 1998]                                  | ✓              | ~ <sup>b</sup> | ×                |
| DPP / submodular coverage [Ye et al., 2023, Ghulyani et al., 2026]             | ✓              | ~ <sup>b</sup> | ×                |
| Directed-information covering [Huang, 2025]                                    | ✓              | ~ <sup>f</sup> | ×                |
| Shapley / influence valuation [Xie et al., 2024]                               | ×              | ✓              | ×                |
| Expected information gain / BED [Choudhury et al., 2025]                       | ×              | ~              | ×                |
| Listwise LLM reranker [Sun et al., 2023]                                       | ~ <sup>c</sup> | ✓              | × <sup>c</sup>   |
| Utility scores, learned or LLM-specific [Zhang et al., 2025, Dai et al., 2025] | ✓ <sup>a</sup> | ×              | ×                |
| Conditional passage utility [Jain and Garimella, 2025]                         | ✓ <sup>a</sup> | ~ <sup>d</sup> | ×                |
| Conformal context filtering [Chakraborty et al., 2025]                         | ✓              | ×              | ×                |
| Learned set selection [Jiang et al., 2026, Fan et al., 2026]                   | ~ <sup>c</sup> | ✓              | × <sup>c</sup>   |
| Skyline over heuristic scores [Zhu et al., 2026]                               | ✓              | ×              | ~ <sup>e</sup>   |
| <b>Verified front + routing (this paper)</b>                                   | ×              | ✓              | ✓                |

Table 2: The design space of context selection. ✓ = yes, × = no, ~ = partial. <sup>a</sup>amortized: trained once on the model’s signal, cheap at query time, but still a scalar. <sup>b</sup>symmetric redundancy over embedding geometry, not directional anti-monotonicity. <sup>c</sup>query-conditioned and interaction-aware, but per-query (not precomputable) and with no dominance guarantee. <sup>d</sup>positive complementarity only, a passage made useful by another, not directional harm. <sup>e</sup>keeps the non-dominated set, but over heuristic per-item scores rather than task value. <sup>f</sup>genuinely *directional*, but the relation is redundancy between chunks (does  $i$  predict  $j$  to within  $\gamma$  bits?), computed offline and query-agnostically, not harm to a task’s verified value. Only the last row represents incomparability over task value, at the cost of direct probing; the “cheap *and* partial-order-faithful” cell is empty, which is this paper’s central open problem.

## 2.4 The design space: no cheap method represents the partial order

Table 2 places the methods above on three axes: whether the method is *cheap* (precomputable, or amortizable to a near-retrieval query-time cost), whether it models *interaction between sources* rather than scoring each in isolation, and whether it can represent *incomparability*. The pattern is uniform. Every method cheap enough to run at retrieval scale emits a final per-item number, however that number is produced, and therefore lives in a total order; every method that genuinely models interaction is either symmetric redundancy over embedding geometry, carries no dominance guarantee, or is as costly as direct probing. No prior row is both cheap and partial-order-faithful; that empty cell is the open problem this paper isolates, and we occupy the last row, paying probe cost for the guarantee.

## 3 Formalization

§2 placed the contribution against prior work; the next four sections make it precise, beginning here with the objects. We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task’s latent requirement. This section fixes the objects and, crucially, separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which is the object every empirical quantity in this paper actually measures. Keeping these distinct is what makes the bridge from theorem to experiment rigorous.



### 3.1 Tasks, latent requirements, and context sources

A *task*  $T$  fixes a latent requirement  $S_T$ , the object a deterministic verifier  $v_T$  checks (e.g. the intended program behaviour). We take  $S_T$  to be a *vector of coordinates*

$$S = (S_1, S_2, \dots, S_m) \in \mathcal{S} = \mathcal{S}_1 \times \dots \times \mathcal{S}_m, \quad (1)$$

where each coordinate  $S_j$  encodes one logically independent requirement (in our instantiation,  $S_1$  is the private call-contract convention and  $S_2$  is the private wire-encoding convention; see §7). Modelling  $S$  as a product of coordinates, rather than an opaque atom, is the technical device that makes the experiment model well posed for fixed-string sources; we return to this in §3.2 and Lemma 1.

**Definition 1** (Context source as an experiment). A *context source*  $W$  is a Markov kernel  $\kappa_W : \mathcal{S} \rightarrow \Delta(\mathcal{W})$ , i.e. a statistical experiment indexed by the latent parameter  $S$ , emitting a signal  $w \sim \kappa_W(\cdot | S)$  in a signal alphabet  $\mathcal{W}$  (the source’s text). The experiment is identified with its kernel; the sample space is  $\mathcal{W}$  and the observed datum is the supplied text  $w$ .

**Definition 2** (Decision rules and PASS). A *decision rule* is a (possibly randomized) map  $\delta : (T, w) \mapsto \hat{S}$  producing a candidate solution. Utility is the binary verifier  $u(\hat{S}, S) = v_T(\hat{S}) \in \{0, 1\}$ . Two decision rules matter:

- (i) the *model rule*  $\delta_M$  realized by a fixed language model  $M$  at a fixed decoding configuration  $\theta$  (model weights and sampling parameters; see §3.3). Its realized pass rate on task  $T$  under source  $W$  is

$$\text{PASS}_M(W, T) := \mathbb{E}[v_T(\delta_M(T, w))], \quad w \sim \kappa_W(\cdot | S_T); \quad (2)$$

- (ii) the *Bayes rule*  $\delta_T^*$ , the (utility-optimal) decision rule for task  $T$  given a prior over  $S$ , achieving the optimal value  $V^*(W, T) := \sup_{\delta} \mathbb{E}[v_T(\delta(T, w))]$ .

The distinction in Definition 2 is load-bearing. Blackwell’s theorem (§4) is a statement about  $V^*$  (the optimal user of an experiment); every number we report,  $\text{PASS}_M$ , the deficiency estimator of §6, the interference statistic of §6.3, and the entire six-model program of §7, is a statement about the single fixed rule  $\delta_M$ . We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

### 3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*: a snippet stating the private call contract ( $W_1$ ), or the private wire-encoding invariant ( $W_2$ ). A naive reading treats  $\kappa_W(\cdot | S)$  as a constant kernel  $\delta_{w_0}$  independent of  $S$ . That reading is degenerate: a constant kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial  $\sigma$ -algebra, which would make any garbling argument vacuous. We avoid this by modelling each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

**Definition 3** (Projection sources). A source  $W$  that reveals the coordinates indexed by a set  $A \subseteq \{1, \dots, m\}$  is the deterministic kernel

$$\kappa_W(\cdot | S) = \delta_{\pi_A(S)}, \quad \pi_A(S) := (S_j)_{j \in A}, \quad (3)$$

i.e. supplying  $W$  lets a decision-maker condition on  $\pi_A(S)$  and on nothing outside  $A$ . Write  $\mathcal{F}_W := \sigma(\pi_A)$  for the sub- $\sigma$ -algebra of  $\mathcal{S}$  generated by the projection. In our instantiation  $W_1$  reveals coordinate  $S_1$  (the call contract,  $A_1 = \{1\}$ ),  $W_2$  reveals  $S_2$  (the wire invariant,  $A_2 = \{2\}$ ), and the superset  $W_{1+} = W_2 ++ W_1$  reveals both ( $A_{1+} = \{1, 2\}$ ).

Under Definition 3 a fixed string is *not* an uninformative constant: it is a deterministic experiment whose informativeness is exactly the resolving power of its  $\sigma$ -algebra  $\mathcal{F}_W$ . The signal alphabet is the range of the projection, and the text “varies with the task” precisely because the realized coordinate values  $\pi_A(S_T)$  differ across task instances  $T$ . This is the well-posed reading that makes the comparison of §4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [Torgersen, 1991].

### 3.3 The conditioning parameter $\theta$

The model rule  $\delta_M$  carries an implicit parameter  $\theta$ : the model’s weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write  $I(S; W \mid \theta)$  for the model-relative value of a private source, the information  $W$  supplies *beyond what  $\theta$  already encodes*. When  $\theta$  already “knows” a convention (the model can guess it without the source),  $W$ ’s marginal value collapses; when the convention is genuinely unknown to  $\theta$ ,  $W$  is maximally valuable. This  $\theta$ -dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in §7, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by  $\theta$ ; we suppress it except where the model-relativity is the point.

## 4 The Blackwell order over context sources

With sources modeled as experiments (§3), the classical comparison of experiments applies to them directly. We state the order, record that it is partial (the regime our experiments target), and, because those experiments measure a fixed model rather than the idealized decision-maker the classical theorem assumes, build an explicit bridge between its two registers.

### 4.1 The order and its two registers

**Definition 4** (Garbling order).  $W_1 \succeq_B W_2$  ( $W_1$  *Blackwell-dominates*  $W_2$ ) iff there is a Markov kernel (garbling)  $G : \mathcal{W}_1 \rightarrow \Delta(\mathcal{W}_2)$  with  $\kappa_{W_2} = G \circ \kappa_{W_1}$ ; i.e.  $W_2$ ’s signal is a noised, post-processed version of  $W_1$ ’s, computable from  $W_1$ ’s signal alone without further access to  $S$ .

The classical content of the order is the Blackwell–Sherman–Stein theorem, which we state with hypotheses for a self-contained treatment.

**Theorem 1** (Blackwell–Sherman–Stein; universal dominance for the optimal rule). *Let  $W_1, W_2$  be experiments on a common parameter space  $\mathcal{S}$  with standard Borel signal spaces. The following are equivalent:*

- (i)  $W_1 \succeq_B W_2$  (Definition 4);
- (ii) *for every prior  $\mu$  on  $\mathcal{S}$ , every finite decision space, and every bounded loss  $\ell$ , the Bayes risk under  $W_1$  is no larger than under  $W_2$ ; equivalently the optimal value satisfies  $V^*(W_1, T) \geq V^*(W_2, T)$  for every task  $T$  with bounded utility.*

*Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii).*

This is the classical Blackwell–Sherman–Stein theorem [Blackwell, 1953, Torgersen, 1991], which we use as a black box. The only specialization is reading “decision problem” as a task with bounded verifier utility  $v_T \in \{0, 1\}$  and “experiment” as a context source (Definition 1), which is faithful because the kernels of Definition 3 are standard Borel.

**Corollary 1** (Universal context dominance, C1). *If  $W_1 \succeq_B W_2$  then, for the optimal user, supplying  $W_1$  is at least as good as supplying  $W_2$  on every task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information,  $\mathcal{V}$ -usable information) provides.*

*Remark 1* (The order is partial; no lattice).  $\succeq_B$  is a genuine partial order: there exist pairs  $W_1, W_2$  with neither  $W_1 \succeq_B W_2$  nor  $W_2 \succeq_B W_1$ , and the order does not form a lattice [Torgersen, 1991]. Incomparability is exactly the regime our experiments target.

## 4.2 Deterministic sources: incomparability via $\sigma$ -algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated  $\sigma$ -algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

**Lemma 1** (Refinement characterization for projection sources). *Let  $W_A, W_B$  be projection sources revealing coordinate sets  $A, B$  (Definition 3), with coordinates  $S_1, \dots, S_m$  mutually independent under some full-support prior and each  $S_j$  non-degenerate. Then:*

- (a)  $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$ ;
- (b) *consequently  $W_A$  and  $W_B$  are Blackwell-incomparable iff  $A \not\subseteq B$  and  $B \not\subseteq A$ , i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in §7: from the wire format ( $W_2$ , revealing  $S_2$ ) one cannot manufacture the private arg-order / required-keyword / exception-type facts ( $S_1$ ), and from the call contract ( $W_1$ , revealing  $S_1$ ) one cannot manufacture the cents-scale / check-digit facts ( $S_2$ ). With  $A_1 = \{1\}$ ,  $A_2 = \{2\}$  neither set contains the other, so  $W_1, W_2$  are Blackwell-incomparable by construction (and  $W_{1+}$ , revealing  $\{1, 2\}$ , dominates both). This guarantees the *theoretical* incomparability the experiments then exhibit empirically.

## 4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns  $V^*$ . Our experiments measure  $\text{PASS}_M$  for a single fixed, generally sub-optimal rule  $\delta_M$ . The two are not interchangeable: an inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a  $\delta_M$ -restricted order and state precisely what it does and does not inherit from Theorem 1.

**Definition 5** ( $\delta_M$ -restricted,  $\mathcal{D}$ -restricted order). Fix the model rule  $\delta_M$  and a finite task set  $\mathcal{D}$ . Say  $W_1 \succeq_{M, \mathcal{D}} W_2$  iff  $\text{PASS}_M(W_1, T) \geq \text{PASS}_M(W_2, T)$  for every  $T \in \mathcal{D}$ . Call  $W_1, W_2$  ( $M, \mathcal{D}$ )-incomparable iff  $W_1 \not\succeq_{M, \mathcal{D}} W_2$  and  $W_2 \not\succeq_{M, \mathcal{D}} W_1$ , i.e. each strictly out-performs the other on some task in  $\mathcal{D}$ .

$\succeq_{M, \mathcal{D}}$  is the order our verification measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite  $\mathcal{D}$ ); within those restrictions it is still a genuine, *for-all-tasks-in- $\mathcal{D}$*  dominance statement, not a scalar.

**Lemma 2** (One-way bridge under coordinate separation). *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

- (a) Let  $W_1, W_2$  be projection sources revealing coordinate sets  $A_1, A_2$  (Definition 3), and let  $T_a, T_b \in \mathcal{D}$  be tasks whose verifier-decisive coordinates lie in  $A_1 \setminus A_2$  and  $A_2 \setminus A_1$  respectively, with guessing caps  $g_a := V^*(W_2, T_a)$  and  $g_b := V^*(W_1, T_b)$ , the optimal values achievable without the decisive coordinate. If  $\text{PASS}_M(W_1, T_a) > g_a$  and  $\text{PASS}_M(W_2, T_b) > g_b$ , then  $W_1, W_2$  are Blackwell-incomparable.
- (b) The converse fails in general: Blackwell incomparability concerns  $V^*$ , and a fixed sub-optimal  $\delta_M$  may fail to realize the optimal-value crossover on  $\mathcal{D}$ . It holds under the value-attainment hypothesis (Assumption 1).

The separation hypothesis in (a) is necessary, and our own measurements witness the necessity:  $(W_1, W_{1+})$  is  $(M, \mathcal{D})$ -incomparable on the measured cells ( $W_1$  wins the API cells,  $W_{1+}$  the encoding cell) yet Blackwell-comparable by construction ( $W_{1+} \succeq_B W_1$ ), precisely because the deprived-side premise fails there,  $W_{1+}$  contains  $W_1$ 's coordinate, so the API-side cap is not small. A bare  $\text{PASS}_M$  crossover, without the separation hypothesis, witnesses only the sub-optimality of  $\delta_M$  (Theorem 3).

Intuitively: a fixed-rule crossover proves structural incomparability only when each winning source reveals a coordinate the losing source lacks; otherwise the crossover may witness nothing more than the rule's own incoherence.

**Assumption 1** (Value attainment on the witnessing tasks). There exist tasks  $T_a, T_b$  and a margin  $\gamma > 0$  such that  $\delta_M$  attains  $\text{PASS}_M(W_1, T_a) - \text{PASS}_M(W_2, T_a) \geq \gamma$  and  $\text{PASS}_M(W_2, T_b) - \text{PASS}_M(W_1, T_b) \geq \gamma$ ; i.e. the fixed model actually exploits each source's private coordinate on the task that requires it. Under Assumption 1 with  $\{T_a, T_b\} \subseteq \mathcal{D}$ ,  $W_1$  and  $W_2$  are  $(M, \mathcal{D})$ -incomparable.

Lemma 2 is the keystone. In the direction the empirical program needs, it says: a verified  $(M, \mathcal{D})$ -crossover on tasks whose decisive coordinates are *separated* across the two sources, with the deprived side's optimal value capped below the measured pass rates, implies genuine Blackwell incomparability. Our construction is designed to satisfy the hypothesis (each cell's decisive private coordinate is revealed by exactly one source, Lemma 1), and its observable signature, deprived-arm pass rates flooring at the guessing cap ( $W_2$  at 0% on the API cells,  $W_1$  at 0% on the encoding cell, none near 0), is verified per cell in §7; where a model's prior partially recovers a convention (the  $\theta$ -dependence of §3.3), the cap premise weakens and we do not verify on those cells. The reverse implication, predicting the realized crossover from the abstract theorem, requires Assumption 1, which our task design enforces and our measurements verify.

## 5 The incomparability theorem

§4 showed the order is partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim into a clean theorem for query-independent source scores  $\varphi(W)$  and a separate, explicitly empirical statement for query-conditioned scores  $\varphi(W, T)$  (the relevance score practitioners actually use); the two must not be conflated.

### 5.1 Query-independent scores: the theorem

**Theorem 2** (No query-independent scalar ranks incomparable sources, C1). *Let  $W_1, W_2$  be Blackwell-incomparable. Let  $\varphi : \{\text{sources}\} \rightarrow \mathbb{R}$  be any query-independent scalar score (a single number per source: relevance to a fixed corpus, mutual information  $I(S; W)$ ,  $\mathcal{V}$ -usable information, perplexity-gap, expected information gain under a fixed prior). Then there exist tasks  $T_a, T_b$  such that the ranking of  $\{W_1, W_2\}$  induced by  $\varphi$  disagrees with the optimal-value ranking on at least one of  $T_a, T_b$ ; and, under Assumption 1, with the realized- $\text{PASS}_M$  ranking on at least one of them.*

*Proof.* By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, “not  $W_1 \succeq_B W_2$ ” yields a task  $T_b$  (a prior plus bounded utility) with  $V^*(W_2, T_b) > V^*(W_1, T_b)$ , and “not  $W_2 \succeq_B W_1$ ” yields a task  $T_a$  with  $V^*(W_1, T_a) > V^*(W_2, T_a)$ ; both inequalities are *strict*, so ties do not arise. A query-independent  $\varphi$  assigns the single ordered pair  $(\varphi(W_1), \varphi(W_2))$ , fixing one total order of  $\{W_1, W_2\}$  *independent of the task*. That fixed order agrees with the optimal-value order on at most one of  $T_a, T_b$  (they point opposite ways), hence is violated on the other. The  $\text{PASS}_M$  conclusion follows by replacing  $V^*$  with  $\text{PASS}_M$  via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin  $\gamma$ .  $\square$

*Remark 2* (Scope). Theorem 2 rules out exactly the class of *source-level* scalars: any  $\varphi$  that is a fixed number per source, regardless of task.  $\mathcal{V}$ -usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1).

*Remark 3* (Machine-checked). Theorem 2 and Proposition 2 are formalized and machine-checked in Lean 4 (`lean/ContextSelection.lean` in the artifact repository, as `no_faithful_query_independent_score` and `topical_score_misranks_trap`). The formalization takes no dependency on Mathlib and checks against a bare `lean` binary; the order on the score codomain is passed explicitly rather than drawn from a typeclass hierarchy, which records the fact that the obstruction is *totality of the induced ranking alone*, not any property of  $\mathbb{R}$ . Every scalar in the excluded class therefore falls under the one theorem. A `Demo` namespace instantiates both results on the paper’s own configuration, so the impossibility statements are not vacuous, and `#print axioms` reports that both central theorems depend on *no* axioms: the arguments are constructive, using neither choice nor excluded middle. What is *not* formalized is stated as plainly: the Blackwell–Sherman–Stein theorem remains a cited classical black box (Theorem 1), and the statistical procedures of §6 are claims about measurements rather than theorems.

**Proposition 1** ( $\mathcal{V}$ -usable information is Blackwell-monotone). *Let  $\mathcal{V}$ -usable information  $I_{\mathcal{V}}(W \rightarrow S)$  be defined, after Ethayarajh et al. [2022], as the reduction in expected predictive Bayes risk of the best predictor in a fixed family  $\mathcal{V}$  when given the signal  $w$  versus the null signal. If  $W_1 \succeq_B W_2$  and the predictor family  $\mathcal{V}$  is closed under pre-composition with measurable garblings, then  $I_{\mathcal{V}}(W_1 \rightarrow S) \geq I_{\mathcal{V}}(W_2 \rightarrow S)$ . Hence  $I_{\mathcal{V}}$  is a Blackwell-monotone scalar: it respects  $\succeq_B$  but collapses it to one real number.*

*Remark 4* (Scope of Proposition 1). The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour,  $\mathcal{V}$ -usable information, and conjecture, rather than prove, the analogue for mutual information, perplexity-gap, and fixed-prior EIG, each of which is a data-processing-monotone functional of the same kernels.

## 5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score  $\varphi(W, T)$ , e.g. the lexical or embedding relevance of a source to the task prompt, which is what retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order. Theorem 2 does *not* cover it. We make two precise statements instead.

**Proposition 2** (Constrained impossibility for feature-measurable scores). *Suppose  $\varphi(W, T)$  is measurable with respect to a fixed feature map of the pair  $(W, T)$  that is topical: it depends on  $W, T$*

only through a feature map that is invariant to the verifier-decisive coordinate of  $S_T$ . Let  $T_c$  be a task whose decisive coordinate is supplied by  $W_1$ , so realized utility ranks  $W_1 \succ W_2$  on  $T_c$ . **(a)** If  $W_1$  and  $W_2$  are topically indistinguishable on  $T_c$ , then  $\varphi$  assigns them equal scores and cannot rank  $W_1$  strictly above  $W_2$ . **(b)** If in addition  $\varphi(\cdot, T_c)$  is strictly increasing in lexical overlap and  $T_c$ 's surface lexicon is saturated with  $W_2$ 's vocabulary, then  $\varphi(W_2, T_c) > \varphi(W_1, T_c)$ , a strict mis-rank. Either way no topical query-conditioned score ranks  $\{W_1, W_2\}$  correctly on all tasks.

*Proof.* For (a), topicality makes  $\varphi(\cdot, T_c)$  a function of the topical features alone, so equal features force equal scores, while faithfulness on  $T_c$  would require a strict gap. For (b), the added strict monotonicity in overlap turns the saturation hypothesis into  $\varphi(W_2, T_c) > \varphi(W_1, T_c)$ ; utility favours  $W_1$  because  $T_c$ 's PASS requires  $W_1$ 's coordinate, so the two orders disagree. Part (a) is the statement machine-checked in Lean (Remark 3); (b) adds the monotonicity hypothesis that the informal reading of “saturated lexicon” presumes, and without which no strict inequality follows.  $\square$

Proposition 2 is the strongest *general* statement we can make about query-conditioned scores: it holds for any score in the topical (lexical-feature) class, but not for an arbitrary learned reranker that could, in principle, read the private coordinate. We therefore treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use. In §7 the `trap_store_wire` task realizes  $T_c$ : bag-of-words cosine gives  $\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$  (ranking  $W_2 \succ W_1$ ), yet realized PASS ranks  $W_1$  at 96%–100% versus  $W_2$  at 0–10% across all six models. The same mis-rank is realized by representative open-weight learned rankers: two dense bi-encoders (BGE, E5) and a cross-encoder reranker all prefer  $W_2$  on the trap (§7.4.2, Table 4), so the topical class is not a lexical strawman: its failure mode is shared by the ranker families retrieval systems actually deploy. The same failure is documented in the wild: language-model re-rankers are steered by lexical similarity across NQ, LitQA2, and DRUID [Hagström et al., 2025], and passages that are semantically close but logically irrelevant defeat retrievers and rerankers at scale, with an evidence-reasoning agent, a listwise reader, as the remedy [Chen et al., 2026], which is the boundary Proposition 2 draws. We also measure the boundary from the other side: an LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (30/30). This is the behaviour Proposition 2 predicts, its hypothesis is topicality (invariance to the verifier-decisive coordinate), and a ranker that reads the coordinate is simply outside the impossibility class.

## 6 The decision-restricted deficiency surrogate and its uniform verification

The impossibility of §5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free verification that makes each comparison auditable on a live model.

### 6.1 The surrogate and its correct name

Exact garblings over text cannot be computed, and  $\succeq_B$  is only partial, so instead we measure the fixed-rule order of Definition 5. Fix the model rule  $\delta_M$  and a finite task set  $\mathcal{D}$ , and define the *decision-restricted value-deficiency*, the worst amount by which  $W_1$  trails  $W_2$  across the tasks in  $\mathcal{D}$ :

$$\hat{\delta}_{\mathcal{D}}(W_1, W_2) := \max\left(0, \sup_{T \in \mathcal{D}} [\text{PASS}_M(W_2, T) - \text{PASS}_M(W_1, T)]\right), \quad (4)$$

estimated per cell by  $\text{PASS}_M(W, T) \approx k_{W,T}/n$  from  $n$  completions. Then  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  verifies empirical dominance  $W_1 \succeq_{M, \mathcal{D}} W_2$ , and mutual positivity  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$  verifies  $(M, \mathcal{D})$ -incomparability.

*Remark 5* (This is *not* Le Cam deficiency). The Le Cam deficiency  $\delta(\mathcal{E}, \mathcal{F}) = \inf_G \sup_{\text{loss}} [R_{G \circ \mathcal{E}} - R_{\mathcal{F}}]$  minimizes over garblings  $G$  and takes a supremum over a class of losses [Le Cam, 1964, 1986, Torgersen, 1991]. The quantity in (4) performs no garbling minimization and uses a single fixed rule  $\delta_M$  over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates  $\hat{\delta}_{\mathcal{D}}$  to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (4) the *decision-restricted value-deficiency* and use “Le Cam deficiency” only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative:  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  realizes the for-all-tasks-in- $\mathcal{D}$  dominance of Definition 5, the measurable shadow of Corollary 1.

## 6.2 The uniform verification

The surrogate (4) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

**Proposition 3** (Uniform incomparability verification). *Within each cell  $(W, T)$  draw  $n$  i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let  $[L_{W,T}, U_{W,T}]$  be the exact (two-sided) Clopper–Pearson interval [Clopper and Pearson, 1934] for  $p_{W,T} = \text{PASS}_M(W, T)$  at level  $\alpha = \eta/(2|\mathcal{D}|)$ . By the Bonferroni/union bound [Bonferroni, 1936] over the  $2|\mathcal{D}|$  proportions  $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$ , all  $2|\mathcal{D}|$  intervals hold simultaneously with probability at least  $1 - \eta$ . On that event,*

$$L_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [L_{W_2,T} - U_{W_1,T}] \quad (5)$$

*is a  $(1 - \eta)$  lower confidence bound on  $\sup_{T \in \mathcal{D}} [p_{W_2,T} - p_{W_1,T}]$ . Hence  $L_{\mathcal{D}}(W_1, W_2) > 0$  verifies  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$  at confidence  $\geq 1 - \eta$ . Both directions  $L_{\mathcal{D}}(W_1, W_2)$  and  $L_{\mathcal{D}}(W_2, W_1)$  read off the same  $2|\mathcal{D}|$  intervals, so the joint incomparability claim  $\{\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0\}$  holds at  $\geq 1 - \eta$  with no additional correction. The matching quantity  $U_{\mathcal{D}}(W_1, W_2) := \max_{T \in \mathcal{D}} [U_{W_2,T} - L_{W_1,T}]$  is, on the same event, a  $(1 - \eta)$  upper confidence bound on the same supremum; the dominance control uses it to verify  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$  up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the verification selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

**Assumption 2** (Sampling regime). The  $n$  outcomes within a cell are i.i.d. Bernoulli draws of  $v_T(\delta_M(T, w))$ . Every draw is a single completion (plus deterministic code extraction) from an independent call with no shared state, so the draws are manifestly independent and the binomial model applies directly; we report the decoding configuration in §7. Where draws are near-deterministic ( $p \in \{0, 1\}$  stark cells), Clopper–Pearson collapses and the verification is conservative.

## 6.3 The interference verification (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source  $W_{1+}$  (revealing strictly more coordinates than  $W_1$ ;

Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T := \text{PASS}_M(W_{1+}, T) - \text{PASS}_M(W_1, T) - \text{PASS}_M(W_2, T) + \text{PASS}_M(\text{none}, T), \quad (6)$$

the second-difference (interaction) of adding  $W_2$  on top of  $W_1$ . We write the interaction as  $\Psi_T$  (rather than a symbol  $I$  that would collide with mutual information  $I(S; W \mid \theta)$ ). A verified  $\Delta_T < 0$  on a cell whose  $W_1$  arm is at ceiling is the localized statement that “more strictly-more-informative context hurts”;  $\Psi_T$  is reported alongside it as the interaction term only (Remark 6).

**Proposition 4** (Interference verification). *Fix a target task  $T^* \in \mathcal{D}$  chosen before unblinding, and let  $\Delta_{T^*} := \text{PASS}_M(W_{1+}, T^*) - \text{PASS}_M(W_1, T^*)$  be the harm contrast. With exact Clopper–Pearson endpoints  $\text{CP}_{\text{lo}}, \text{CP}_{\text{hi}}$  at level  $\alpha = \eta/2$  on the two arms  $\{W_{1+}, W_1\}$ ,*

$$\overline{\Delta}_{T^*} := \text{CP}_{\text{hi}}(p_{W_{1+}}) - \text{CP}_{\text{lo}}(p_{W_1}) \quad (7)$$

*is a  $(1 - \eta)$  upper confidence bound on  $\Delta_{T^*}$ , and  $\overline{\Delta}_{T^*} \leq -\tau$  verifies  $\Delta_{T^*} \leq -\tau$  at confidence  $\geq 1 - \eta$ . If instead the harmful cell is selected from  $|\mathcal{D}|$  candidates after unblinding, use  $\alpha = \eta/(2|\mathcal{D}|)$ ; this is the same selection correction the verification applies to the deficiency surrogate (Remark 7), and omitting it would repeat the multiplicity error we criticize. Because  $\Delta$  reads only the two arms it contrasts, the bound is insensitive to the **none** baseline; we still report that baseline per cell, because guessability bears on whether the source is genuinely private and hence on the incomparability reading (§3.3).*

*Remark 6* ( $\Psi$  is interaction, not harm). The second difference must not be read as a harm measure. Writing  $f$  for the set function  $A \mapsto \text{PASS}_M(W_A, T)$ , the inequality  $\Psi_T \leq 0$  is precisely the submodularity condition for the pair  $\{W_1, W_2\}$ , so a negative  $\Psi_T$  is what diminishing returns look like and is consistent with  $f$  being monotone submodular. It does not imply that adding  $W_2$  to  $W_1$  hurt: taking  $\text{PASS}(\text{none}) = 0$  and  $\text{PASS}(W_1) = \text{PASS}(W_2) = \text{PASS}(W_{1+}) = 1$  gives  $\Psi_T = -1$  with no degradation whatever. What the collapse violates is *monotonicity*,  $\Delta_T < 0$ , which is the quantity we verify. In our data the two coincide in direction, and the contrast bound is tighter than the four-term one because it spends its confidence budget on two intervals rather than four.

## 6.4 Anti-monotonicity as information non-monotonicity of $\delta_M$

A verified  $\Delta_{T^*} < 0$  on a ceiling- $W_1$  cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

**Theorem 3** (Information non-monotonicity of the model rule). *Let  $W_{1+}$  reveal a superset of  $W_1$ ’s coordinates, so  $W_{1+} \succeq_B W_1$  (Lemma 1). Any Blackwell-coherent rule, one whose value is monotone under  $\succeq_B$ , as the Bayes rule  $\delta^*$  is by Theorem 1, satisfies  $\text{value}(W_{1+}, T) \geq \text{value}(W_1, T)$  for every task  $T$ . If a fixed model rule  $\delta_M$  has a task  $T^*$  with  $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$ , then  $\delta_M$  is not Blackwell-coherent: it violates monotonicity under the garbling order on  $T^*$ . In particular no monotone scalar context-value score can predict  $\delta_M$ ’s behaviour on  $T^*$ .*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent, which is precisely the gap Lemma 2(b) warned cannot be closed without value attainment. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.



## 6.5 Family-wise error across the verified claims

Each verification above controls error *marginally* at  $\eta$  for its own claim (incomparability via Proposition 3; interference via Proposition 4). When several claims are read off the *same*  $n$ -sample run at the same  $\eta$ , incomparability (both directions, on one shared event  $E$ ), one-directional dominance, and the interference cells, the conjunction over distinct claims is *not* jointly controlled at  $\eta$  unless the budget is split across claims. We state results as marginally verified per claim and, where a joint statement is wanted, allocate  $\eta$  across the  $C$  verified claims (e.g.  $\eta/C$  each) so the global family-wise error is  $\leq \eta$ . The incomparability pair is the one exception that is genuinely joint at  $\eta$  on a single event, as proved in Proposition 3; all other cross-claim conjunctions on a shared run are reported as marginal.

The two verifications have different effective- $n$  requirements: incomparability is carried by stark 0/1 cells whose intervals are tight, whereas a harm contrast on a milder collapse spends more of the budget. This is why the four-term interaction bound left Opus short of the margin while the two-arm contrast clears it (§7.4.3); the power analysis is in Appendix A.

## 7 Experiments

With the order (§4), its impossibility consequence (§5), and the verification that measures it (§6) in hand, we now instantiate the theory on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (§7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every verified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (§7.4), so the verification is auditable rather than asserted.

### 7.1 Design: two incomparable context sources

We define two *private*, *unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any “ledger” task, any lexical, embedding, mutual-information, or  $\mathcal{V}$ -usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

$W_1$ , **API call contract.** `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry id  $\geq 1$ ; raises `ledger.PostError` (not `ValueError/KeyError`) on failure.

$W_2$ , **wire-encoding invariant.** Amounts serialize to `<sign><minor-unit-digits>#<check>`, where `check` = (sum of digits) mod 10; e.g. `$12.30`  $\mapsto$  `" +1230#6"` and `-$0.07`  $\mapsto$  `" -7#7"`. Decoding rejects a wrong check digit.

$W_{1+}$ , **literal superset.**  $W_{1+} = W_2 ++ W_1$ , the concatenation of  $W_2$ ’s signal with  $W_1$ ’s. As a deterministic source revealing both coordinates, it Blackwell-dominates both  $W_1$  and  $W_2$  ( $W_{1+} \succeq_B W_1$  and  $W_{1+} \succeq_B W_2$ ); it is the anti-monotonicity / dominance probe. We also use  $W_{1+}^{\text{rev}} = W_1 ++ W_2$  ( $W_1$  first) as an order control.

**Why incomparable (garbling argument).** The argument is simple. From  $W_2$ ’s wire format you cannot recover  $W_1$ ’s private argument order, its required `memo` keyword, or its error type; from  $W_1$ ’s contract you cannot recover  $W_2$ ’s cents scale, its `#` separator, or its mod-10 check. Each source states a fact the other leaves out, so neither is a blurred copy of the other, and  $W_1, W_2$  are incomparable. This is exactly the coordinate picture of §3:  $W_1$  and  $W_2$  reveal different, non-overlapping facts about  $S$  (Lemma 1).

**Confound defense: the double crossover.** The key pattern is a *double crossover*, measured against a `none` baseline (no note supplied):  $W_1$  raises the API tasks but *not* the encoding task, while  $W_2$  raises the encoding task but *not* the API tasks. If encoding were simply “harder,”  $W_1$  would fail to help it and  $W_2$  would still have to raise it from the same floor. In the clean carrying cells the deprived arms floor at or near the model’s guessing cap, so the two task families are equally hard on their own and any difference comes from the matching private fact, not from one family being harder; when a model prior leaks a convention (Table 7), we report that leakage explicitly and do not use those cells for claims that require a near-zero baseline.

## 7.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four  $W_1$ -decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two  $W_2$ -decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

**Verified battery.** The verified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the verified battery. The four-cell selection was made to retain at least one  $W_1$ -decisive, one  $W_2$ -decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (§7.4.1).

**The relevance trap.** The trap `trap_store_wire` is the confound-proof form of Theorem 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks  $W_2 > W_1$ ; but its graded requirement is the  $W_1$  call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks  $W_1 > W_2$ . This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine  $\varphi_{\text{lex}}(W, T)$  between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in §7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

## 7.3 Methodology

**Models and vendors.** We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6. Nothing in this

paper is trained or fine-tuned: all six are fixed production systems accessed through their public APIs, so there are no training or validation curves to report; the released measurement record, per-cell PASS tables, exact intervals, and raw run logs, is the complete empirical object.

**Harness and arms.** The harness (`harness/measure_blackwell.py`; released with all run logs at <https://github.com/abilous-ti/blackwell-llm-context>) is Python-standard-library only (zero external dependencies); it computes  $\hat{\delta}_{\mathcal{D}}$  in both directions, the uniform Clopper–Pearson/Bonferroni verification, the interference verification, and the lexical relevance score, and it implements the launchers for the Anthropic CLI, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms  $\{\text{none}, W_1, W_2, W_{1+}\}$ . PASS is measured over  $n$  completions per cell.

**One uniform run mode.** All six models are queried the same way: a single completion per attempt (the Anthropic models via `claude -p` with a one-turn budget and no tools, the others through an OpenAI-style API), from which we extract the candidate code and grade it. Because the run mode is identical across vendors, a difference between models is a fact about the model, not about how it was run. This is a deliberately conservative probe of whether a model uses supplied context: it removes the agentic-exploration pathway, so the model must use what it is given in one pass.

**What “verified” means, in plain terms.** Every claim labeled *verified* is statistically significant in the standard sense and then corrected for multiplicity: PASS rates are estimated from  $n = 40$  independent completions per cell, uncertainty is carried by exact binomial (Clopper–Pearson) confidence intervals with no distributional assumptions, and each family of claims is corrected by a union bound, so the statements *within* a family (all incomparability cells, or the arms of one interference contrast) hold jointly with probability at least 90%. We do not claim joint control across families drawn from the same run; that limitation is stated where it bites (§6.5). For the effect sizes at stake, near-0% versus near-100%, the  $n = 40$  intervals stay separated by a wide margin even after correction. Results whose intervals did not separate are labeled *reproduced but not verified* and are never used as evidence; the protocol reports its own insufficiency rather than hiding it.

**Verification parameters.** All verifications use confidence budget  $\eta = 0.10$ . Incomparability uses exact Clopper–Pearson intervals at per-cell level  $\alpha = \eta/(2|\mathcal{D}|)$  over the  $2|\mathcal{D}|$  proportions  $\{p_{W_1,T}, p_{W_2,T}\}_{T \in \mathcal{D}}$ , so by the union bound all hold jointly with probability  $\geq 1 - \eta$  and both directions read off the same intervals. The harm verification uses  $\alpha = \eta/2$  on the two arms  $\{W_{1+}, W_1\}$  and the threshold  $\tau = 0.30$ , verifying  $\Delta_T \leq -\tau$  via  $\overline{\Delta}_T$ ; the four-term  $\Psi_T$  is reported descriptively only (Remark 6). We report each verification as marginally controlled at  $\eta$ ; we do not claim joint family-wise control across the incomparability, dominance, and interference claims drawn from a single run (§6.5, §8.4). The intervals are degenerate at  $p = 0$  and  $p = 1$ , which is why the stark 100%/0% cells verify at small  $n$  while cells with intermediate proportions need a larger  $n$  for the same margin.

**Why these constants.** Both constants were fixed before the runs and neither is load-bearing.  $\eta = 0.10$  is a conventional 90% family level; the headline separations are near-0% against near-100%, so the verified conclusions are not sensitive to tightening it.  $\tau = 0.30$  is an a priori materiality threshold: to count, interference must destroy at least 30 PASS points, well above the interval widths at  $n = 40$ , so noise cannot clear it. The data show the threshold doing real work in both directions.

Table 3: Consolidated 6-model / 4-vendor verdict. All six models are run the same way, a single API call per attempt at  $n = 40$  per cell (§7.3). “Anti-mono. verified” counts, of the two API cells tested per model, those on which the harm contrast’s upper bound clears  $-\tau = -0.30$  at  $\eta = 0.10$ . “Argorder none” is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers.

| Claim                               | Haiku     | Sonnet    | Opus      | GPT-5.5 | DeepSeek | Kimi-K2.6 |
|-------------------------------------|-----------|-----------|-----------|---------|----------|-----------|
| Vendor                              | Anthropic | Anthropic | Anthropic | OpenAI  | DeepSeek | Moonshot  |
| $n$ per cell                        | 40        | 40        | 40        | 40      | 40       | 40        |
| Incomp. verified ( $\geq 90\%$ )    | ✓         | ✓         | ✓         | ✓       | ✓        | ✓         |
| $L_{\mathcal{D}}(W_1, W_2)$         | +0.76     | +0.72     | +0.72     | +0.76   | +0.35    | +0.68     |
| $L_{\mathcal{D}}(W_2, W_1)$         | +0.76     | +0.76     | +0.76     | +0.76   | +0.72    | +0.76     |
| Relevance mis-rank (trap)           | ✓         | ✓         | ✓         | ✓       | ✓        | ✓         |
| Anti-mono. verified (# cells, of 2) | 1         | 2         | 2         | 2       | 2        | 2         |
| <code>api_argorder</code> none      | 2%        | 0%        | 40%       | 75%     | 2%       | 45%       |

The verified collapses clear it with upper confidence bounds from  $-0.37$  to  $-0.82$  (Table 5), and the length-matched padding control clears it at  $-0.63$ . Two real effects fail it and are reported as unverified rather than folded in: Haiku’s `api_post_ok` cell (upper  $-0.06$ ; a ceiling the raw-failure audit traces to an omitted import line, not to the contract, §7.5.1), which still falls short at  $-0.23$  once the import is neutralized, and the second-pair contrast (upper  $-0.296$ , sign-only; §7.5). A threshold that genuine effects fail is evidence it was not tuned to pass the data.

**Sampling regime.** The verification models each cell as  $n$  Bernoulli PASS draws (Assumption 2); the four arms within a cell are separate completions, and each of the  $n$  draws is an independent call with no shared state, so the draws are manifestly independent. We report the decoding parameters, seeding policy, and retry policy in Appendix C.

**Spend.** The record reflects roughly \$120 of Anthropic spend on the single-shot Haiku, Sonnet, and Opus runs (itemized in Appendix C) plus Azure quota for the non-Anthropic models; the Azure API returns no cost field, so this figure is not the total cost across all six models.

## 7.4 Results

We report four findings: incomparability (§7.4.1), the relevance mis-rank (§7.4.2), anti-monotonicity (§7.4.3), and the model-relative value of private context (§7.4.4). Table 3 consolidates the per-model verdicts; Tables 5 and 7 give the underlying quantities.

### 7.4.1 Incomparability verified on all six models

On every model the empirical deficiency is mutually positive,  $\hat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$  and  $\hat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$ , and the uniform lower bounds  $L_{\mathcal{D}}$  in both directions clear zero, verifying empirical incomparability at  $\geq 90\%$  (Table 3). The crossover is stark:  $W_1$  solves the API and trap cells and fails encoding, while  $W_2$  solves encoding and fails the API and trap cells. For Haiku at  $n = 40$  the carrying cells are `api_argorder` (none 2% /  $W_1$  100% /  $W_2$  0%) and `enc_amount` (none 0% /  $W_1$  0% /  $W_2$  100%), giving  $L_{\mathcal{D}} = +0.762 / +0.762$ , far clear of zero.

Table 4: The ranker spectrum on the probe tasks. Each cell is the ranker’s top pick between  $\{W_1, W_2\}$ ; “PASS” is the realized-utility ranking from the verified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools  $n = 30$  judgments per task (unanimous, both presentation orders).  $\times$  marks a mis-rank. The cross-encoder’s `enc_amount` pick is a near-tie ( $-0.279$  vs  $-0.305$ ).

| Ranker                    | Family           | trap (PASS: $W_1$ ) | api_post_ok (PASS: $W_1$ ) | enc_amount (PA |
|---------------------------|------------------|---------------------|----------------------------|----------------|
| bag-of-words cosine       | lexical          | $W_2 \times$        | $W_1$                      | $W_2$          |
| bge-small-en-v1.5         | dense bi-encoder | $W_2 \times$        | $W_1$                      | $W_2$          |
| e5-small-v2               | dense bi-encoder | $W_2 \times$        | $W_1$                      | $W_2$          |
| ms-marco-MiniLM-L-6-v2    | cross-encoder    | $W_2 \times$        | $W_1$                      | $W_1 \times$   |
| Haiku listwise ( $n=30$ ) | LLM listwise     | $W_1$               | $W_1$                      | $W_2$          |

Because incomparability is a two-directional claim, both  $L_{\mathcal{D}}$  bounds must be positive; the smaller of the two is the binding one. The two carrying cells are one API cell and one encoding cell, so the effective support of the surrogate is two cells. We therefore describe  $\mathcal{D}$  as a fixed adversarial four-cell battery rather than a sampled task distribution: the verification is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition replicated across six models.

#### 7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks  $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$ , so it prefers  $W_2$ ; yet realized PASS ranks  $W_1 \gg W_2$  on all six models (Haiku 90% vs. 0%; Sonnet, Opus, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%). On the honest cells relevance *agrees* with PASS, on the API cells relevance ranks  $W_1 > W_2$  and so does PASS, and on the encoding cell relevance ranks  $W_2 > W_1$  and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The **none** floor on the trap is 0% on all six models, and  $W_2$  scores 0% there too, confirming both that the convention is unguessable and that encoding knowledge is useless on this task.

**The ranker spectrum: dense retrievers and a cross-encoder fall for the trap; a listwise reranker escapes it.** To test whether the mis-rank extends beyond bag-of-words, we scored  $\{W_1, W_2\}$  against the probe tasks with representative open-weight rankers, two dense bi-encoders (bge-small-en-v1.5, e5-small-v2) and a cross-encoder reranker (ms-marco-MiniLM-L-6-v2), and with a RankGPT-style LLM *listwise* reranker (Haiku single-shot, seeing the task and *both* sources; both presentation orders;  $n = 30$  judgments per task). Table 4 reports the induced rankings. **Every feature-based ranker mis-ranks the trap:** both dense bi-encoders prefer  $W_2$  on the trap while ranking both honest cells correctly, textbook topical behaviour, and the cross-encoder prefers  $W_2$  on the trap with maximal confidence (logits  $-5.63$  for  $W_1$  vs  $+3.24$  for  $W_2$ ) while being near-tied (and narrowly wrong) on the honest encoding cell. The LLM listwise reranker, which can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere (30/30 per task, both orders). This is exactly the boundary Proposition 2 draws: the impossibility hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense, and cross-encoder families that satisfy it while sparing the one ranker that reads content. The practical corollary is that escaping the topical class costs a model call per candidate, a probe of the same kind the deficiency verification is built from, not a free scalar.

*Idealized Blackwell order (optimal user)*      *Verified empirical order (fixed rule  $\delta_M$ , Haiku  $n=40$ )*

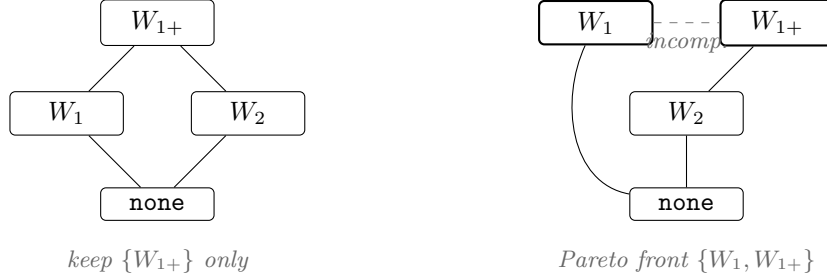


Figure 1: The paper’s thesis in one picture, on the measured  $\mathcal{C} = \{W_1, W_2, W_{1+}\}$  with  $W_{1+} = W_2 ++ W_1$ . **Left:** the idealized Blackwell order for the optimal user;  $W_{1+}$  reveals both coordinates, so “more signal is better” collapses the choice to the superset. **Right:** the verified empirical order for the fixed model rule (Haiku,  $n=40$ ):  $W_1$  beats  $W_{1+}$  on `api_argorder` (100% vs. 5%, anti-monotonicity, Table 5) while  $W_{1+}$  beats  $W_1$  on the encoding cell, so the two are empirically incomparable (dashed) and the correct output is the two-element Pareto front, which no scalar top-1 rule can represent.

### 7.4.3 Anti-monotonicity: a verified superset that hurts

The superset  $W_{1+}$ , which literally *contains*  $W_1$ ’s signal, sharply degrades the API cells relative to  $W_1$  alone. The per-task harm contrast  $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$  is verified negative ( $\Delta_T \leq -\tau$ ,  $\tau = 0.30$ ) on eleven of the twelve API cells, two per model, and so on at least one cell for all six models (Table 5); the mildest verified collapse is Opus at 100%  $\rightarrow$  38% with upper bound  $-0.37$ , and the one cell that does not clear is Haiku’s `api_post_ok`, whose  $W_1$  arm is depressed by an unrelated import slip (§7.5.1). Ten of the eleven survive a Bonferroni correction across the two cells within each model ( $\alpha = \eta/4$ ); DeepSeek’s `api_argorder` is the one that falls just short, at  $-0.298$ . This is a verified empirical falsification of Blackwell superset-dominance on live models (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Table 7 gives the same collapse arm by arm for every model; Haiku’s  $W_1$  reads 52% on `api_post_ok` only because about half its replies omit the `import ledger` line while calling the contract correctly (§7.5.1), and its verified collapse is on `api_argorder`, 100%  $\rightarrow$  5%.

**The empirical order inverts the idealized one.** The idealized Blackwell order on  $\mathcal{C} = \{W_1, W_2, W_{1+}\}$  collapses the choice to the superset:  $W_{1+}$  reveals both coordinates, so  $W_{1+} \succeq_B W_1$  and  $W_{1+} \succeq_B W_2$ , and “more signal is better” keeps  $\{W_{1+}\}$  alone. The verified empirical relation for the fixed rule is the opposite shape:  $W_1$  beats  $W_{1+}$  on the API cells,  $W_{1+}$  beats  $W_1$  on the encoding cell,  $W_{1+}$  dominates  $W_2$  (while  $W_1$  and  $W_2$  are the verified incomparable pair), so the correct empirical summary is the two-element Pareto front  $\{W_1, W_{1+}\}$ , an object no scalar top-1 rule can represent (Figure 1). This reads directly off the verified  $n=40$  Haiku cells; the selection rule it implies, keep the verified non-dominated set rather than force the superset, follows directly.

**Ceilings differ, so magnitudes are not directly comparable.** Table 5 reports, per model, which cell carried the verification and its collapse, alongside the  $W_1$  ceiling: DeepSeek’s  $W_1$  reaches only 75% on `api_post_ok` (an honest capability difference), so its  $\Delta_T = -0.75$  starts from a lower point and is not directly magnitude-comparable to Sonnet’s 100%  $\rightarrow$  2%.

Table 5: Anti-monotonicity verification on the carrying cell per model.  $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$  is the harm contrast;  $\bar{\Delta}_T$  is its  $\eta/2$  Clopper–Pearson upper bound over the two arms, and the cell is verified harmful when  $\bar{\Delta}_T \leq -0.30$ . The  $W_1 \rightarrow W_{1+}$  column reports the PASS collapse. Unlike the four-term interaction  $\Psi_T$ , the contrast does not read the `none` arm, so a leaky baseline cannot slacken it. <sup>†</sup>Control row: the same cell with the omitted-`import` slip neutralized in the verifier (§7.5.1); its interference is verified in sign but not at the  $\tau = 0.30$  margin.

| Model           | Carrying cell                                              | $W_1 \rightarrow W_{1+}$ | $\Delta_T$ | $\bar{\Delta}_T$ | Verified                    |
|-----------------|------------------------------------------------------------|--------------------------|------------|------------------|-----------------------------|
| Haiku-4.5       | <code>api_argorder</code>                                  | 100% $\rightarrow$ 5%    | -0.95      | -0.74            | ✓                           |
| Haiku-4.5       | <code>api_post_ok</code>                                   | 52% $\rightarrow$ 15%    | -0.38      | -0.06            | sign only (import omission) |
| Haiku-4.5       | <code>api_post_ok</code> , import-neutralized <sup>†</sup> | 100% $\rightarrow$ 52%   | -0.47      | -0.23            | sign only <sup>†</sup>      |
| Sonnet-4.6      | <code>api_post_ok</code>                                   | 100% $\rightarrow$ 2%    | -0.97      | -0.78            | ✓                           |
| Sonnet-4.6      | <code>api_argorder</code>                                  | 100% $\rightarrow$ 0%    | -1.00      | -0.82            | ✓                           |
| GPT-5.5         | <code>api_post_ok</code>                                   | 100% $\rightarrow$ 0%    | -1.00      | -0.82            | ✓                           |
| GPT-5.5         | <code>api_argorder</code>                                  | 100% $\rightarrow$ 0%    | -1.00      | -0.82            | ✓                           |
| DeepSeek-V4-Pro | <code>api_post_ok</code>                                   | 75% $\rightarrow$ 0%     | -0.75      | -0.50            | ✓                           |
| DeepSeek-V4-Pro | <code>api_argorder</code>                                  | 68% $\rightarrow$ 5%     | -0.62      | -0.34            | ✓                           |
| Kimi-K2.6       | <code>api_post_ok</code>                                   | 100% $\rightarrow$ 0%    | -1.00      | -0.82            | ✓                           |
| Kimi-K2.6       | <code>api_argorder</code>                                  | 100% $\rightarrow$ 0%    | -1.00      | -0.82            | ✓                           |
| Opus-4.8        | <code>api_post_ok</code>                                   | 100% $\rightarrow$ 38%   | -0.62      | -0.37            | ✓                           |
| Opus-4.8        | <code>api_argorder</code>                                  | 100% $\rightarrow$ 15%   | -0.85      | -0.61            | ✓                           |

**Position is not the cause.** The collapse is not a lost-in-the-middle or recency artifact. The length-matched padding control (Appendix B) puts filler in the same slot  $W_2$  occupies inside  $W_{1+}$ , with  $W_1$  last in both, and leaves  $W_1$  at ceiling; holding position fixed and changing only the content is what turns a harmless arm into a harmful one, so the cause is the added signal, not where it sits.

**Runtime fingerprint (a descriptive correlate).** The harmful  $W_{1+}$  arm consistently emits more output tokens than  $W_1$  across models. We report this as a reproducible correlate of the harmful arm, not a mechanism: more output on the failing arm could be effect rather than cause.

**Robust, but not monotone in capability.** The harm is large and robust across the Anthropic capability ladder but is *not* ordered by capability. On the two API cells the  $W_1 \rightarrow W_{1+}$  collapse is Haiku 52%/100%  $\rightarrow$  15%/5%, Sonnet 100%  $\rightarrow$  2%/0%, and Opus 100%  $\rightarrow$  38%/15%: the middle model (Sonnet) shows the starkest collapse and the strongest (Opus) the mildest. An earlier two-point reading suggested the harm “intensifies with strength”; the third point falsifies that, and we claim only robustness across the ladder, not a capability trend.

#### 7.4.4 The model-relative value of private context

The “private” conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The `none`-arm baseline on `api_argorder` forms a clean spectrum: Sonnet 0%, Haiku and DeepSeek 2%, Opus 40%, Kimi 45%, GPT-5.5 75% (Table 3). So the value of the private context, the model-relative quantity  $I(S; W \mid \theta)$  of §3.3, varies continuously across models, and it tracks a vendor’s training rather than raw capability: DeepSeek, a different vendor, has a tight prior like the smaller Claude models, while GPT-5.5’s prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%), so a “stronger-model-knows-more” account does

not fit. Utility-centric retrieval reaches the same conclusion from the other side, the same passage carrying different utility for different models [Zhang et al., 2025]; what the spectrum adds is the verified per-model floor. Within Anthropic the ordering is not by capability either: Sonnet guesses least (0%) while the stronger Opus guesses more (40%). The verified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok none` = 5%, `trap` = 0%).

**Auditable per-cell record.** The full per-arm PASS record for every carrying cell is Table 7 in Appendix B, so each verification can be reconstructed from the reported counts.

## 7.5 Generality and controls (summary)

Four follow-up experiments address three threat axes (source-pair generality, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i) A second incomparable pair, different domain:** a private `cache-module` contract versus a key-normalization invariant is again verified incomparable on Haiku ( $L_D = +0.774 / +0.496$ ), with one-directional dominance fully verified at  $n = 80$ ; its harm contrast is verified in sign ( $\Delta_T < 0$ , upper bound  $-0.296$ ) but not at the  $\tau = 0.30$  margin, because the 80%  $V_1$  ceiling keeps the bound wide. **(ii) Length-matched padding:** filler character-matched to  $W_{1+}$  leaves  $W_1$  at ceiling (`api_argorder` 100% vs  $W_{1+}$  10%; harm verified, upper  $-0.63$ ), attributing the collapse to the added signal, not token count. **(iii) Routing instruction:** a one-line convention-routing note recovers most of the collapse (2%  $\rightarrow$  80%, 22%  $\rightarrow$  68%; both separated from  $W_{1+}$ ) while remaining directionally below  $W_1$  alone on the carrying cell; writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(iv) Structured context:** wrapping the same two conventions in labeled XML sections *without* any routing statement mitigates only weakly (2%  $\rightarrow$  42% on the carrying cell, versus 80% with the routing note and 100% for  $W_1$  alone; verified against the raw superset, still far below  $W_1$ ), so segmentation alone does not cure the interference, the operative variable is routing knowledge. The mitigation hierarchy, raw superset  $<$  structure  $<$  routing  $<$  dropping the dominated signal, is itself a finding: it locates the failure in convention-to-task routing rather than in convention segmentation.

### 7.5.1 Raw-failure audit: what Haiku’s `api_post_ok` cell measures

One cell in Table 7 is anomalous: Haiku’s  $W_1$  arm scores 52% on `api_post_ok` while scoring 100% on the harder `api_argorder` under the same source, and the pattern is stable across five independent Haiku runs (45–57%). Because the published harness does not retain raw completions, we re-drew the cell keeping every reply, the extracted code, and the verifier’s stderr (`results/diag/`, `harness/diag/`). The failures are uniform. In 12/12 draws the reply calls the private contract exactly as  $W_1$  specifies, `ledger.post(acct, cents, memo=note)`, and every failure is the single omitted line `import ledger`: PASS coincides with the presence of that line (5/5 against 7/7), and on the control cell `api_argorder` the same model writes the import in 8/8. The cause is prompt wording, not the contract: this task’s prompt names the call as `ledger.post`, which Haiku reads as already in scope. An import-neutralized control then re-measures all four arms at  $n=40$  with a verifier that binds `ledger` into the solution’s namespace when the reply omitted the import and is otherwise identical: `none` 0/40 and  $W_2$  0/40 under both verifiers (neutralization leaks nothing),  $W_1$  rises from 23/40 to 40/40, and  $W_{1+}$  from 8/40 to 21/40. The superset collapse on this cell is therefore real and has the mechanism the paper describes: in the retained  $W_{1+}$  replies the model applies  $W_2$ ’s wire encoding to a plain-cents amount and posts `'+500#5'` in place of 500. With



| Model           | Best fixed source | Topical top-1 | Concatenate ( $W_{1+}$ ) | Front + routing |
|-----------------|-------------------|---------------|--------------------------|-----------------|
| Haiku-4.5       | 60.5              | 63.0          | 53.0                     | <b>85.5</b>     |
| Sonnet-4.6      | 75.0              | 74.5          | 36.8                     | <b>99.5</b>     |
| Opus-4.8        | 75.0              | 74.5          | 59.5                     | <b>99.5</b>     |
| GPT-5.5         | 75.0              | 75.0          | 50.0                     | <b>100.0</b>    |
| DeepSeek-V4-Pro | 60.3              | 52.8          | 42.0                     | <b>77.3</b>     |
| Kimi-K2.6       | 75.0              | 73.8          | 48.0                     | <b>98.8</b>     |
| Mean            | 70.1              | 68.9          | 48.2                     | <b>93.4</b>     |

Table 6: Mean PASS (%) over the four verified cells under four selection policies, per model, computed from the measured per-cell rates of Table 7. The front-and-routing column is the per-cell maximum of  $W_1$  and  $W_2$  and is therefore the ceiling of *any* selection policy on these cells; its remaining gap to 100% is model capability, not selection error.

the import neutralized the harm contrast is  $\Delta_T = -0.47$  with upper bound  $-0.23$ , verified in sign but not at the  $\tau = 0.30$  margin (Table 5). We keep the original-protocol numbers in every table, since the verifier is identical across models and the other five models include the import; the audit changes the reading of one cell, not its measurement. Haiku’s verified claims rest on `api_argorder` and are unaffected.

## 7.6 The engineering consequence: four selection policies, one measured table

The formal results reduce to a comparison an engineer can act on. On the measured pair there are four things a selection layer can do, and every one of them was measured (Table 7); Table 6 scores them per model as mean PASS over the four verified cells. *Best fixed source* grants the entire query-independent-scalar class its best case: any such scalar induces one total order, hence one top-1 reused for every task, and we let it pick the oracle-best source ( $W_1$  for every model). *Topical top-1* re-scores per task with topical relevance, choosing  $W_1$  on the API cells and  $W_2$  on encoding, but also  $W_2$  on the trap, where every measured topical ranker puts the wrong source first (§7.4.2). *Concatenate both* refuses to choose and submits  $W_{1+}$ . *Front + routing* keeps the verified non-dominated set  $\{W_1, W_2\}$  and routes by task family, attaining the per-cell maximum of the two sources; this is what the partial order licenses, and its price is exactly one probe’s worth of knowledge, namely which source is decisive for which family.

Three structural facts, not the magnitudes, are the point. First, the two scalar policies fail on *different* cells, the fixed source can never carry encoding, topical top-1 forfeits the trap, yet land within about one point of the same aggregate: the cap is the policy *class*, since any total order must abandon at least one cell, not the choice of scorer (Theorem 2, Proposition 2). Second, the policy that refuses to choose is the *worst* of the four: anti-monotonicity is the dominant cost, not a tie-breaking subtlety (§7.4.3). Third, the front column shows what selection alone can recover: its shortfall from 100% is not selection error. For DeepSeek (68–75% band) it is the model’s own capability limit; for Haiku the apparent 52% ceiling on `api_post_ok` is not even that. A raw-failure audit (§7.5.1) shows every Haiku reply on that cell calls the private contract correctly and the deficit is an omitted `import` line; with the import neutralized,  $W_1$  scores 40/40. One honest caveat: the four cells were constructed to exhibit the phenomenon, so the aggregate magnitudes are diagnostic of the failure structure, not an estimate of production-traffic gains; what transfers is the structure, each cheap policy forfeits an entire cell class, and no total-order policy can keep all four.

**What follows for practice.** Five statements survive the caveats above and can be acted on directly. (i) One number per source is structurally lossy: when two sources each carry a decisive fact, any fixed score drops one of them, and no threshold repairs it (Theorem 2). (ii) Do not cache a source’s *value* across task families; cache retrieval representations instead, because the value is model- and task-relative (§7.4.4). (iii) Adding correct, on-topic text to a working prompt can break it, by as much as  $100\% \rightarrow 0\%$ , so a regression after enlarging the context is a candidate anti-monotonicity collapse before it is a model failure (§7.4.3). (iv) When two candidates win on different tasks, keep both and route by task rather than forcing a global top-1: the front attains the per-cell maximum of its members (Table 6). (v) A cheap topical ranker fails exactly where the surface lexicon points at the wrong source, which is where a listwise reader earns its cost (Proposition 2). Each is measurable on any stack with the released harness.

## 8 Discussion

With the evidence in hand, we step back to what it licenses. Our results recast a piece of practitioner folklore, “relevance does not transfer across tasks”, as a structural fact about context sources. When two sources are Blackwell-incomparable, the order over them is partial, and any scalar score, by imposing a total order, must mis-rank them on some task. The empirical program does not merely illustrate this; it instantiates a single hand-built incomparable pair (a private API-call contract  $W_1$  versus a private wire-encoding invariant  $W_2$  on the same `ledger` module) and shows that the resulting double crossover, the relevance mis-rank, and an anti-monotonicity collapse all reproduce across six models and four vendors. We discuss what the evidence does and does not license, and where the formalization is a model rather than a proof of the realized object.

### 8.1 Scope of the claims

Three limits of scope govern everything above. **(i) Two registers.** The theorems are about the ideal user of a source; every number we measure is the pass rate of one fixed, non-optimal rule  $\delta_M$ . Lemma 2 connects the two in the direction we need: a verified crossover on tasks where each source supplies its own decisive fact implies genuine Blackwell incomparability, and the reverse needs an extra assumption (Assumption 1) that our task design builds in and our measurements confirm. So we read the result as a statement about the fixed model on a fixed battery that *earns* the structural conclusion through the bridge rather than assuming it. **(ii) The estimator’s name.**  $\hat{\delta}_{\mathcal{D}}$  does no minimization over garblings and ranges over a fixed set of tasks; it is a one-sided *value-deficiency surrogate*, not a Le Cam deficiency, and the two are not interchangeable for a fixed rule (Remark 5). What is rigorous is the verification, which we regard as the most robust part and which stands whether or not the deficiency analogy holds. **(iii) What the verification covers.** It is correct because it pays for picking the worst task with exact intervals and a union bound; its support is small (one API task and one encoding task carry the pair-1 result), each claim is controlled on its own at level  $\eta$  rather than all at once (§6.5), and we call  $\mathcal{D}$  a fixed adversarial battery, not a task distribution.

### 8.2 Anti-monotonicity is real, verified, but not capability-ordered

The harm object is the direct contrast  $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$ , which measures whether appending the (irrelevant-to-this-task)  $W_2$  to a sufficient  $W_1$  degrades performance (the second difference  $\Psi_T$  is an interaction term, not a harm measure; Remark 6). We verify  $\Delta_T \leq -0.30$  on at least one cell for all six models (Table 5), with collapses as stark as  $100\% \rightarrow 0\%$  (upper bound

−0.82) on Sonnet, GPT-5.5 and Kimi. This is a verified empirical falsification of superset-dominance for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone scalar context-value score can represent.

Two cautions attach to it. The harm is robust across the capability ladder but not ordered by it (§7.4.3), so we claim no capability trend. And which cell carries the verification is still a choice: we fix it per model before unblinding rather than picking the starkest afterwards, and cells differ in their  $W_1$  ceiling, so magnitudes are not comparable across models even though each binary verdict stands on its own.

**Will scale restore Blackwell coherence?** Three points on one vendor’s ladder cannot answer this, and we do not extrapolate; what the framework contributes is the instrument, since coherence is now a verifiable property of any fixed rule and “ $\Delta_T$  versus scale” is a measurable trajectory. Whatever that curve shows, the central claim is unaffected: anti-monotonicity is a coherence failure of the *rule* and could in principle vanish with scale, but incomparability is a property of the *sources*, and mutually non-refining signals stay incomparable however capable the reader becomes.

### 8.3 Mechanism: the runtime fingerprint is a correlate, not an explanation

Every collapse is accompanied by output-token inflation on the harmful arm. On the two API collapse cells  $W_{1+}$  emits between +93% and +520% more output tokens than  $W_1$  (Haiku +115%/ +163%; Sonnet +520%/ +473%; GPT-5.5 +281%/ +274%; DeepSeek +93%/ +166%; Kimi +203%/ +249%; the re-measured Opus  $W_{1+}$  arm carries no token record). One might read this as “verbose thrashing” causing the failure, but the relationship is correlational and uncontrolled: on the *still-passing* encoding cell the same superset arm ranges from −73% to +114% across the same models, with no consistent direction, which undercuts a clean “verbosity causes collapse” account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

The claim structure is deliberately mechanism-agnostic: Theorem 3 verifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls already constrain the hypothesis space from outside, the harm is not position (order control), not token count (padding control), and is largely curable by explicit convention routing (instruction ablation), a signature consistent with mis-binding rather than capacity exhaustion. That the one-line routing note recovers most of the collapse while labeled XML segmentation of the same content recovers only a minority (2% → 42% against → 80%) locates the operative variable in convention-to-task routing rather than formatting; this is an interpretation, not a finding, and every verified claim is independent of it. An activation-level account is out of reach here, since four of the six models are closed-weight and the phenomenon is cross-vendor; the direct route is to reproduce the collapse on an open-weight model with the same harness and instrument the contrasting arms, downstream of rather than prerequisite to the behavioral claim. Distractor interference in RAG has been localized to a sparse feature subspace and corrected there [Giannetti et al., 2026]; ours differs in cause, the added source is true and needed rather than a distractor, and it is not a knowledge conflict in the sense of Xu et al. [2024], since nothing in  $W_1$  and  $W_2$  contradicts.

### 8.4 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

**From diagnosis to selection.** The constructive selection rule the partial order implies, keep the verified non-dominated set and cover what it leaves uncovered, is left for future work; here we note

only its cost. Building the verified order on  $m$  candidates costs  $m \cdot |\mathcal{D}| \cdot n$  verifier-gated calls, curation scale, not corpus scale; scaling verified selection to retrieval over  $10^6$  passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens. Keeping a non-dominated set instead of a top- $k$  prefix already exists as an operator, the skyline retrieval step of Zhu et al. [2026], and learned set selectors optimize evidence sets end to end [Jiang et al., 2026, Fan et al., 2026]; what the partial order adds is the object the set is taken over, verified task value rather than heuristic per-item scores, which is exactly what makes the selection safe under anti-monotonicity.

**Source-pair generality.** Incomparability and dominance hold for two hand-built pairs in two domains; the cross-vendor record is single-pair. Pair-2 harm is verified in sign ( $\bar{\Delta}_T = -0.296$ ) but not at the  $\tau = 0.30$  margin, because its 80%  $V_1$  ceiling keeps the bound wide; pair-2’s relevance trap was an honest null (its prompt was not lexically  $V_2$ -leaning). A third domain and a vendor-general strong mis-rank remain inexpensive next steps.

**Cross-vendor magnitudes are still only suggestive.** All six models are now run the same way (single-shot,  $n = 40$ ), so the run mode no longer confounds the comparison. What remains is the ordinary caution that different vendors’ models differ in ways we do not control (training data, size, tokenizer), so the *magnitude* comparisons (the value spectrum, the  $\Delta_T$  ladder) should be read as suggestive; the binary within-model claims do not depend on any cross-vendor comparison.

**The model is an idealization, twice.**  $\hat{\delta}_{\mathcal{D}}$  verifies dominance and incomparability for a given model on a fixed battery, not the universal Blackwell order. And the projection construction models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured  $\theta$ -dependence is exactly such leakage), so the construction is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs.

**Prevalence is not measured, and is the decisive next experiment.** Every claim here is existential: it establishes *that* verified incomparability and superset harm occur, on which models, and with which confounds excluded. It does not establish *how often* either occurs in a deployed retrieval stream, and no number in this paper should be read as an estimate of that. The four cells were built to make the structure visible, so their aggregate magnitudes are diagnostic of the failure mode rather than of production traffic. The direct experiment is specified and inexpensive: draw document pairs from a natural corpus, run the same verification battery unchanged, and report the fraction of pairs that verify incomparable and the fraction of superset unions that verify harmful, with the sampling frame stated. We name it as the open question rather than approximate it here, because a prevalence figure computed from hand-built pairs would be circular.

**Sampling and verification regime.** The verification assumes  $n$  i.i.d. Bernoulli PASS draws per cell (Assumption 2), which holds because each draw is an independent single-shot call (protocol in Appendix C); marginal-versus-joint control is discussed in §8.1.

**Ranker spectrum and synergy.** The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker escaping as predicted, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. Positive interference (synergy) is untested by construction:  $W_1, W_2$  are task-disjoint, so a

complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

## 9 Conclusion

Context selection is, at bottom, a partial-order problem, not a scalar one. When two sources are incomparable, no single per-source score, relevance or utility, can rank them correctly for all tasks (Theorem 2), which explains the field’s “relevance does not transfer” folklore and gives utility-centric retrieval its formal boundary. We made the order measurable with a value-deficiency surrogate  $\hat{\delta}_{\mathcal{D}}$  and a distribution-free verification, and we ran the theory on six models across four vendors. On every model we verify ( $\geq 90\%$ ) that two private sources are incomparable; on every model a real relevance score ranks them the wrong way; and on all six we verify that adding a strictly larger source sharply lowers the pass rate (as deep as  $100\% \rightarrow 0\%$ ), which no “more-is-better” scalar can predict. How much a private source is worth turns out to depend on the model: guessability runs from 0% to 75% across vendors and tracks a vendor’s training rather than raw capability.

On scope: the theorems concern the idealized Bayes-optimal user of a source; the realized crossover, anti-monotonicity, and value-of-private-context are properties of *fixed* models measured directly, bridged to the structural claim by Lemma 2. The estimator is a regret-gap surrogate, not a Le Cam deficiency; the verification, which we regard as the most robust contribution, is correct and survives that re-labeling. All six models are run the same way (single-shot,  $n = 40$ ), and the evidence spans two hand-built source pairs in two domains plus a length-matched padding control that attributes the anti-monotonicity collapse to signal content rather than added tokens; the cross-vendor record remains single-pair, cross-vendor magnitudes are only suggestive, and synergy is untested. The practical reading is modest and actionable: selection should be treated as partial-order-aware rather than scalar top- $k$ , the right operational object is a verified deficiency surrogate rather than a relevance scalar, and “the best context” should be expected to be model-relative and only partially ordered. Stated at its sharpest, the impossibility is a theorem about *cheap* selection: per-source scalars (Theorem 2) and topical query-conditioned scores (Proposition 2), not all context selection. The measured spectrum realizes exactly that boundary: lexical cosine, two dense bi-encoders, and a cross-encoder all fall for the trap, while an LLM listwise reranker, which pays per-candidate computation to read the decisive content, escapes it. The hierarchy is computational: scalar scores cannot represent the order, and probes that read content can measure it, at probe cost. The remaining strengthenings, a vendor-general strong mis-rank, larger reranker checkpoints, and a third domain, are the natural next steps, and the experimental record indicates each is cheap to run.

## Author Contributions

Conceptualization, A.B. and O.R.; methodology, A.B. and O.R.; software, A.B.; validation, A.B. and O.R.; formal analysis, A.B. and O.R.; investigation, A.B.; data curation, A.B.; writing—original draft preparation, A.B.; writing—review and editing, A.B. and O.R.; visualization, A.B. All authors have read and agreed to the published version of the manuscript.

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## Institutional Review Board Statement

Not applicable. The study involved no human participants and no animals; all measurements are automated queries to commercial language-model endpoints, graded by executable verifiers.

## Informed Consent Statement

Not applicable.

## Data Availability Statement

All data, code and machine-checked proofs supporting the reported results are openly available at <https://github.com/abilous-ti/blackwell-llm-context>. The repository contains the measurement harness (Python standard library only), every per-model run log and result file cited in Appendix B, the raw-completion diagnostic records of §7.5.1, and the Lean 4 formalization of Remark 3 with the continuous-integration job that re-checks it. No data were withheld; every reported figure is computed from those files by the released scripts.

## Conflicts of Interest

The authors declare no conflict of interest.

## Abbreviations

The following abbreviations are used in this manuscript:

|      |                                                           |
|------|-----------------------------------------------------------|
| BSS  | Blackwell–Sherman–Stein (theorem)                         |
| CP   | Clopper–Pearson (exact binomial confidence interval)      |
| DPP  | Determinantal point process                               |
| EIG  | Expected information gain                                 |
| IR   | Information retrieval                                     |
| LLM  | Large language model                                      |
| MMR  | Maximal marginal relevance                                |
| PASS | Pass rate of a fixed model rule on a verifier-graded task |
| PRP  | Probability ranking principle                             |
| RAG  | Retrieval-augmented generation                            |

## References

- Deniz Akdemir. Le Cam distortion: A decision-theoretic framework for robust transfer learning. *arXiv preprint arXiv:2512.23617*, 2025.
- Pia Alpila. Context selection as approximate sufficiency: Le Cam deficiency and Kolmogorov’s structure function against a fixed decoder, 2026a. Preprint, 25 June 2026; companion essays 10.5281/zenodo.20799732 and 10.5281/zenodo.20800098.
- Pia Alpila. Context selection has been a solved problem since 1951, 2026b. Expository companion to Alpila [2026a].
- Pia Alpila. LM Recall: corpus, experiment code and per-decoder results, 2026c. Software and data artifact.

- Peter Auer, Chao-Kai Chiang, Ronald Ortner, and Madalina Drugan. Pareto front identification from stochastic bandit feedback. In *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 51 of *PMLR*, pages 939–947, 2016.
- David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2): 265–272, 1953. doi: 10.1214/aoms/1177729032.
- Carlo Emilio Bonferroni. Teoria statistica delle classi e calcolo delle probabilità. *Pubblicazioni del R. Istituto Superiore di Scienze Economiche e Commerciali di Firenze*, 8:3–62, 1936.
- Sam Bowyer, Laurence Aitchison, and Desi R. Ivanova. Position: Don’t use the CLT in LLM evals with fewer than a few hundred datapoints. *arXiv preprint arXiv:2503.01747*, 2025. ICML 2025.
- William F. Bradley. Enhancing LLM evaluations: The garbling trick. *arXiv preprint arXiv:2411.01533*, 2024.
- Jaime Carbonell and Jade Goldstein. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 335–336, 1998. doi: 10.1145/290941.291025.
- Debashish Chakraborty, Eugene Yang, Daniel Khashabi, Dawn Lawrie, and Kevin Duh. Principled context engineering for RAG: Statistical guarantees via conformal prediction. *arXiv preprint arXiv:2511.17908*, 2025.
- Kwan Ho Ryan Chan, Yuyan Ge, Edgar Dobriban, Hamed Hassani, and René Vidal. Conformal information pursuit for interactively guiding large language models. *arXiv preprint arXiv:2507.03279*, 2025.
- Haotian Chen, Qingqing Long, Siyu Pu, Xiao Luo, Wei Ju, Meng Xiao, Yuanchun Zhou, Jianghua Zhao, and Xuezhi Wang. DeepEra: A deep evidence reranking agent for scientific retrieval-augmented generated question answering. *arXiv preprint arXiv:2601.16478*, 2026.
- Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, Yizhe Zhang, and Tom Rainforth. BED-LLM: Intelligent information gathering with LLMs and bayesian experimental design. *arXiv preprint arXiv:2508.21184*, 2025. ICLR 2026.
- C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. doi: 10.1093/biomet/26.4.404.
- Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. The power of noise: Redefining retrieval for RAG systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2024.
- Lu Dai, Yijie Xu, Jinhui Ye, Hao Liu, and Hui Xiong. SePer: Measure retrieval utility through the lens of semantic perplexity reduction. In *International Conference on Learning Representations (ICLR)*, 2025.
- Xin Luna Dong, Barna Saha, and Divesh Srivastava. Less is more: Selecting sources wisely for integration. *Proceedings of the VLDB Endowment*, 6(2):37–48, 2012. URL <http://www.vldb.org/pvldb/vol16/p37-dong.pdf>. VLDB 2013.
- Yufeng Du, Minyang Tian, Srikanth Ronanki, Subendhu Rongali, Sravan Bodapati, Aram Galstyan, Azton Wells, Roy Schwartz, Eliu A. Huerta, and Hao Peng. Context length alone hurts LLM performance despite perfect retrieval. *arXiv preprint arXiv:2510.05381*, 2025. Findings of EMNLP 2025.
- A. Dvoretzky, J. Kiefer, and J. Wolfowitz. Asymptotic minimax character of the sample distribution function and of the classical multinomial estimator. *The Annals of Mathematical Statistics*, 27(3):642–669, 1956. doi: 10.1214/aoms/1177728174.

- Kawin Ethayarajh, Yejin Choi, and Swabha Swayamdipta. Understanding dataset difficulty with  $\mathcal{V}$ -usable information. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *PMLR*, pages 5988–6008, 2022.
- Yongqi Fan, Yuxiang Chu, Zhentao Xia, Xiaoyang Chen, Jie Liu, Haijin Liang, Jin Ma, Ben He, Yingfei Sun, Jie Zhai, Dezhi Ye, and Tong Ruan. Rank4Gen: RAG-preference-aligned document set selection and ranking. *arXiv preprint arXiv:2601.11273*, 2026.
- Loïc Fosse, Frederic Bechet, Benoit Favre, Géraldine Damnati, Gwénolé Lecorvé, Maxime Darrin, Philippe Formont, and Pablo Piantanida. Statistical deficiency for task inclusion estimation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Volume 1: Long Papers*, 2025.
- Markus Frohmann, Mahdiyar Alavi, Elizabeth Lingg, and Navid Rekabsaz. Equal ranking quality, different decisions: Training order-consistent LLM scorers. *arXiv preprint arXiv:2608.26762*, 2026.
- Norbert Fuhr. A probability ranking principle for interactive information retrieval. *Information Retrieval*, 11(3):251–265, 2008. doi: 10.1007/s10791-008-9045-0.
- Irena Gao, Percy Liang, and Carlos Guestrin. Model equality testing: Which model is this API serving? In *International Conference on Learning Representations (ICLR)*, 2025.
- Manu Ghulyani, Arunabh Singh, Karan Bharadwaj, Ankit Nath, and Suranjan Goswami. PACMS: Submodular context selection as a pluggable engine for LLM agents. *arXiv preprint arXiv:2606.20047*, 2026.
- Christian Giannetti, Giovanni Trappolini, Nicola Tonellotto, Fabrizio Silvestri, and Pietro Liò. The mechanics of interference: Defusing distractors in RAG via sparse autoencoder interventions. In *Findings of the Association for Computational Linguistics: ACL 2026*, 2026. URL <https://aclanthology.org/2026.findings-acl.583/>.
- Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. REALM: Retrieval-augmented language model pre-training. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- Lovisa Hagström, Ercong Nie, Ruben Halifa, Helmut Schmid, Richard Johansson, and Alexander Junge. Language model re-rankers are fooled by lexical similarities. *arXiv preprint arXiv:2502.17036*, 2025. FEVER 2025.
- Alon Halevy and Jane Dwivedi-Yu. Learnings from data integration for augmented language models. *arXiv preprint arXiv:2304.04576*, 2023.
- Kelly Hong, Anton Troynikov, and Jeff Huber. Context rot: How increasing input tokens impacts LLM performance. Technical report, Chroma, 2025. URL <https://www.trychroma.com/research/context-rot>.
- Hai Huang. Directed information  $\gamma$ -covering: An information-theoretic framework for context engineering. *arXiv preprint arXiv:2510.00079*, 2025.
- Akriti Jain and Aparna Garimella. Modeling contextual passage utility for multihop question answering. In *Proceedings of IJCNLP-AAACL 2025*, 2025.
- Yi Jiang, Sendong Zhao, Jianbo Li, Bairui Hu, Yanrui Du, Haochun Wang, and Bing Qin. OptiSet: Unified optimizing set selection and ranking for retrieval-augmented generation. *arXiv preprint arXiv:2601.05027*, 2026.
- Bowen Jin, Jinsung Yoon, Jiawei Han, and Sercan O. Arik. Long-context LLMs meet RAG: Overcoming challenges for long inputs in RAG. *arXiv preprint arXiv:2410.05983*, 2024.



- Hailey Joren, Jianyi Zhang, Chun-Sung Ferng, Da-Cheng Juan, Ankur Taly, and Cyrus Rashtchian. Sufficient context: A new lens on retrieval augmented generation systems. *arXiv preprint arXiv:2411.06037*, 2024. ICLR 2025.
- Omar Khattab and Matei Zaharia. ColBERT: Efficient and effective passage search via contextualized late interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 39–48, 2020. doi: 10.1145/3397271.3401075.
- Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4): 1419–1455, 1964. doi: 10.1214/aoms/1177700372.
- Lucien Le Cam. *Asymptotic Methods in Statistical Decision Theory*. Springer Series in Statistics. Springer, New York, 1986. doi: 10.1007/978-1-4612-4946-7.
- Will LeVine and Bijan Varjavand. Relevance isn’t all you need: Scaling RAG systems with inference-time compute via multi-criteria reranking. *arXiv preprint arXiv:2504.07104*, 2025. ICLR 2025.
- Shahar Levy, Nir Mazon, Lihi Shalmon, Michael Hassid, and Gabriel Stanovsky. More documents, same length: Isolating the challenge of multiple documents in RAG. *arXiv preprint arXiv:2503.04388*, 2025.
- Ming Liu. More is not always better: Cross-component interference in LLM agent scaffolding. *arXiv preprint arXiv:2605.05716*, 2026.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.
- Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.
- Ikhtiyor Nematov, Tarik Kalai, Elizaveta Kuzmenko, Gabriele Fugagnoli, Dimitris Sacharidis, Katja Hose, and Tomer Sagi. Source attribution in retrieval-augmented generation. In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2025)*, 2025. doi: 10.1007/978-3-032-19099-4\_23.
- Napoleon Paxton. The theorems of dr. david blackwell and their contributions to artificial intelligence. *arXiv preprint arXiv:2604.06621*, 2026.
- Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation. *arXiv preprint arXiv:2605.03534*, 2026.
- Stephen E. Robertson. The probability ranking principle in IR. *Journal of Documentation*, 33(4):294–304, 1977. doi: 10.1108/eb026647.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023a.
- Weijia Shi, Sewon Min, Michihiro Yasunaga, Minjoon Seo, Rich James, Mike Lewis, Luke Zettlemoyer, and Wen-tau Yih. REPLUG: Retrieval-augmented black-box language models. *arXiv preprint arXiv:2301.12652*, 2023b. NAACL 2024.
- Weiwei Sun, Lingyong Yan, Xinyu Ma, Shuaiqiang Wang, Pengjie Ren, Zhumin Chen, Dawei Yin, and Zhaochun Ren. Is ChatGPT good at search? Investigating large language models as re-ranking agents. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Erik N. Torgersen. *Comparison of Statistical Experiments*, volume 36 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991. ISBN 9780521250306. doi: 10.1017/CBO9780521250306.

- Brendan van Rooyen and Robert C. Williamson. Le Cam meets LeCun: Deficiency and generic feature learning. *arXiv preprint arXiv:1402.4884*, 2014.
- M. S. Vinay, Minh-Hao Van, and Xintao Wu. In-context learning demonstration selection via influence analysis. *arXiv preprint arXiv:2402.11750*, 2024.
- Dingzirui Wang, Xuanliang Zhang, Keyan Xu, Qingfu Zhu, Wanxiang Che, and Yang Deng. When does context help? error dynamics of contextual information in large language models. *arXiv preprint arXiv:2602.08294*, 2026.
- Jun Wang and Jianhan Zhu. Portfolio theory of information retrieval. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 115–122, 2009. doi: 10.1145/1571941.1571963.
- Shan Xie, Man Luo, Chadly Daniel Stern, Mengnan Du, and Lu Cheng. DemoShapley: Valuation of demonstrations for in-context learning. *arXiv preprint arXiv:2410.07523*, 2024.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for LLMs: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.
- Jiacheng Ye, Zhiyong Wu, Jiangtao Feng, Tao Yu, and Lingpeng Kong. Compositional exemplars for in-context learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- Yue Yu, Wei Ping, Zihan Liu, Boxin Wang, Jiaxuan You, Chao Zhang, Mohammad Shoeybi, and Bryan Catanzaro. RankRAG: Unifying context ranking with retrieval-augmented generation in LLMs. *arXiv preprint arXiv:2407.02485*, 2024. NeurIPS 2024.
- Xiaowei Yuan, Zhao Yang, Ziyang Huang, Yequan Wang, Siqi Fan, Yiming Ju, Jun Zhao, and Kang Liu. Exploiting contextual knowledge in LLMs through V-usable information based layer enhancement. *arXiv preprint arXiv:2504.15630*, 2025. ACL 2025.
- Hengran Zhang, Keping Bi, Jiafeng Guo, Jiaming Zhang, Shuaiqiang Wang, Dawei Yin, and Xueqi Cheng. LLM-specific utility for retrieval-augmented generation. *arXiv preprint arXiv:2510.11358*, 2025. CIKM 2026.
- Hengran Zhang, Minghao Tang, Keping Bi, and Jiafeng Guo. Beyond relevance: Utility-centric retrieval in the LLM era. *arXiv preprint arXiv:2604.08920*, 2026a. SIGIR 2026 tutorial.
- Zheng Zhang, Cuong C. Nguyen, Kevin Wells, and Gustavo Carneiro. Multi-agent decision making: A Blackwell’s informativeness approach. *arXiv preprint arXiv:2605.06028*, 2026b.
- Guang Zhao, Byung-Jun Yoon, Gilchan Park, Shantenu Jha, Shinjae Yoo, and Xiaoning Qian. Pareto prompt optimization. In *International Conference on Learning Representations (ICLR)*, 2025.
- Shuran Zheng, Xuan Qi, Rui Ray Chen, Yongchan Kwon, and James Zou. Proper dataset valuation by pointwise mutual information. *arXiv preprint arXiv:2405.18253*, 2024.
- Xinyi Zhu, Haoyang Li, Yongqi Zhang, and Lei Chen. Skyline retrieval meets set-cover chunk merging: A cost-effective RAG-sketch for long-context LLM QA. *Proceedings of the ACM on Management of Data*, 4(3):1–27, 2026. doi: 10.1145/3802111.
- Guido Zuccon and Leif Azzopardi. Using the quantum probability ranking principle to rank interdependent documents. In *Advances in Information Retrieval (ECIR 2010)*, LNCS 5993, pages 357–369, 2010. doi: 10.1007/978-3-642-12275-0\_32.

## A Deferred proofs and technical material

### Proof of Lemma 1

*Proof.* (a) For deterministic experiments a garbling  $G$  with  $\kappa_{W_B} = G \circ \kappa_{W_A}$  must compute  $\pi_B(S)$  from  $\pi_A(S)$  alone (no further access to  $S$ ). Such a measurable map exists iff  $\pi_B$  is  $\sigma(\pi_A)$ -measurable, i.e.  $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$ . Under coordinate independence and non-degeneracy,  $\sigma(\pi_B) \subseteq \sigma(\pi_A)$  holds iff every coordinate in  $B$  is among those in  $A$ , i.e.  $B \subseteq A$ : if some  $j \in B \setminus A$ , then  $S_j$  is independent of  $\pi_A(S)$  and non-constant, so it is not  $\sigma(\pi_A)$ -measurable, and no garbling can reconstruct it; conversely if  $B \subseteq A$  the coordinate-deletion map is a (deterministic) garbling. (b) is the contrapositive of (a) applied in both directions.  $\square$

### Proof of Lemma 2

*Proof.* (a) Suppose, for contradiction,  $W_2 \succeq_B W_1$ . By Theorem 1,  $V^*(W_2, T) \geq V^*(W_1, T)$  for every task  $T$ , in particular at  $T_a$ . But  $V^*(W_1, T_a) \geq \text{PASS}_M(W_1, T_a) > g_a = V^*(W_2, T_a)$ , the first inequality because the optimal rule weakly improves on any fixed rule, the second by hypothesis, a contradiction. Hence  $W_2 \not\succeq_B W_1$ . Symmetrically,  $\text{PASS}_M(W_2, T_b) > g_b = V^*(W_1, T_b)$  forbids  $W_1 \succeq_B W_2$ . Neither dominance direction holds, so  $W_1, W_2$  are Blackwell-incomparable. The caps are essential: without them a  $\text{PASS}_M$  crossover is compatible even with Blackwell *equivalence*, since a fixed rule keyed to one signal's surface form can under-perform on the other, exactly the non-coherence phenomenon Theorem 3 verifies for  $(W_1, W_{1+})$ .

(b) Assumption 1 directly yields  $\text{PASS}_M(W_1, T_a) > \text{PASS}_M(W_2, T_a)$  and the reverse at  $T_b$ , which is Definition 5's incomparability; the failure of the unconditional converse is witnessed by any fixed rule that violates monotonicity the optimal value obeys.  $\square$

### Proof of Proposition 1

*Proof.* Let  $G$  be the garbling with  $\kappa_{W_2} = G \circ \kappa_{W_1}$ . Any predictor  $g \in \mathcal{V}$  acting on  $W_2$ 's signal is realized on  $W_1$ 's signal by the composite  $g \circ G$ , which lies in  $\mathcal{V}$  by the closure hypothesis. Therefore the best  $\mathcal{V}$ -predictor under  $W_1$  attains at least the predictive accuracy of the best under  $W_2$ , so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under  $W_1$ . The scalarity is immediate:  $I_{\mathcal{V}}$  returns a single real number per source, so it induces a total order and, by Theorem 2, cannot rank a Blackwell-incomparable pair correctly for all tasks.  $\square$

### Proof of Proposition 3

*Proof.* Let  $E$  be the event that all  $2|\mathcal{D}|$  Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most  $\alpha = \eta/(2|\mathcal{D}|)$  (exact coverage of Clopper–Pearson [Clopper and Pearson, 1934]), so by the union bound  $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$ , giving  $\Pr[E] \geq 1 - \eta$ . On  $E$ , for every  $T$  we have  $p_{W_2, T} \geq L_{W_2, T}$  and  $p_{W_1, T} \leq U_{W_1, T}$ , so  $p_{W_2, T} - p_{W_1, T} \geq L_{W_2, T} - U_{W_1, T}$ ; taking  $\max_T$  on both sides,  $\sup_T [p_{W_2, T} - p_{W_1, T}] \geq L_{\mathcal{D}}(W_1, W_2)$ . Thus on  $E$  (probability  $\geq 1 - \eta$ ) the true sup is at least  $L_{\mathcal{D}}(W_1, W_2)$ , which is the claimed lower confidence bound; if it is positive, the true sup is positive, verifying  $\delta_{\mathcal{D}}(W_1, W_2) > 0$ . The reverse direction uses  $L_{W_1, T}$  and  $U_{W_2, T}$ , which are endpoints of the *same*  $2|\mathcal{D}|$  intervals already included in  $E$ ; therefore both one-sided verifications are valid on the single event  $E$ , and their conjunction holds at  $\geq 1 - \eta$  without a further Bonferroni factor.  $\square$

## Proof of Proposition 4

*Proof.* The four arms are independent Bernoulli samples (separate completions), so on the intersection of the four Clopper–Pearson events, probability  $\geq 1 - 4\alpha = 1 - \eta$  by the union bound, each true proportion lies in its interval. Substituting the coverage-respecting endpoint for each term of (6) (upper endpoints for the positively-signed  $p_{W_{1+}}, p_{\text{none}}$ , lower endpoints for the negatively-signed  $p_{W_1}, p_{W_2}$ ) gives  $\Psi_{T^*} \leq \bar{\Psi}_{T^*}$  on that event. For a post-hoc selected cell, replacing  $\alpha$  by  $\eta/(4|\mathcal{D}|)$  and union-bounding over the  $4|\mathcal{D}|$  endpoints preserves  $\geq 1 - \eta$  simultaneous coverage.  $\square$

## Proof of Theorem 3

*Proof.*  $W_{1+} \succeq_B W_1$  by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)). By Theorem 1 any rule whose value respects  $\succeq_B$  weakly improves under the dominating experiment. The hypothesized strict decrease  $\text{PASS}_M(W_{1+}, T^*) < \text{PASS}_M(W_1, T^*)$  contradicts monotonicity, so  $\delta_M$  is not Blackwell-coherent. A monotone scalar  $\phi$  assigns  $\phi(W_{1+}) \geq \phi(W_1)$  (monotonicity), hence orders  $W_{1+}$  above  $W_1$  and cannot reproduce the realized reversal.  $\square$

## Why a naive DKW bound fails

*Remark 7* (Why a naive DKW bound fails). The Dvoretzky–Kiefer–Wolfowitz inequality [Dvoretzky et al., 1956] bounds the sup-deviation of *one* empirical CDF from *its own* population CDF over a single i.i.d. sample. Our object is different in two ways that DKW does not handle: it is a max over  $|\mathcal{D}|$  *differences of two binomial means across distinct cells*, and the max *selects* the argmax-gap cell. A single-sample sup inequality neither spans distinct cells nor pays for the selection, so its nominal level under-covers a supremum over  $|\mathcal{D}|$  comparisons; the Type-I error grows with  $|\mathcal{D}|$ . The multiplicity must be paid explicitly via the union bound (Proposition 3), not absorbed into a single-sample DKW envelope.

## Relation to LLM-evaluation statistics

*Remark 8* (Relation to LLM-evaluation statistics). The verification sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [Miller, 2024], two-sample tests verifying behavioral differences between API-served models [Gao et al., 2025], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell  $n$  is below a few hundred [Bowyer et al., 2025], which is precisely our regime ( $n \in [20, 80]$  per cell) and precisely what the exact Clopper–Pearson construction provides.

## Power and the role of $n$

Both verifications are driven by how close the arms sit to an endpoint. For a stark cell with empirical proportion at an endpoint ( $\hat{p} \in \{0, 1\}$ ) the Clopper–Pearson interval collapses to a one-sided bound that clears 0 even at small  $n$ , and the incomparability verification (5) is carried by exactly such cells ( $W_1$  at  $\approx 100\%$ ,  $W_2$  at  $0\%$  and vice versa). The harm contrast (7) needs the gap between two arms to exceed  $\tau$  after both intervals are paid for, so a mild collapse from a non-endpoint arm, as in the import-neutralized Haiku control ( $100\% \rightarrow 52\%$ ), is verified in sign but not at  $\tau = 0.30$  at  $n = 40$ ; that cell would clear at  $n \approx 80$ .

Table 7: Per-cell PASS rates (%) for all four verified cells, by arm, all single-shot at  $n = 40$ . Every model was measured on every cell and arm. The encoding rows (`enc_amount`) carry the  $W_2 > W_1$  direction of incomparability: on every model  $W_2$  solves encoding while  $W_1$  fails it, mirroring the API cells where  $W_1$  solves and  $W_2$  fails.

| Model           | Cell                         | none | $W_1$ | $W_2$ | $W_{1+}$ |
|-----------------|------------------------------|------|-------|-------|----------|
| Haiku-4.5       | <code>api_post_ok</code>     | 0    | 52    | 0     | 15       |
| Haiku-4.5       | <code>api_argorder</code>    | 2    | 100   | 0     | 5        |
| Haiku-4.5       | <code>enc_amount</code>      | 0    | 0     | 100   | 100      |
| Haiku-4.5       | <code>trap_store_wire</code> | 0    | 90    | 0     | 92       |
| Sonnet-4.6      | <code>api_post_ok</code>     | 0    | 100   | 0     | 2        |
| Sonnet-4.6      | <code>api_argorder</code>    | 0    | 100   | 0     | 0        |
| Sonnet-4.6      | <code>enc_amount</code>      | 0    | 0     | 98    | 80       |
| Sonnet-4.6      | <code>trap_store_wire</code> | 0    | 100   | 0     | 65       |
| Opus-4.8        | <code>api_post_ok</code>     | 0    | 100   | 0     | 38       |
| Opus-4.8        | <code>api_argorder</code>    | 40   | 100   | 0     | 15       |
| Opus-4.8        | <code>enc_amount</code>      | 0    | 0     | 98    | 85       |
| Opus-4.8        | <code>trap_store_wire</code> | 0    | 100   | 0     | 100      |
| GPT-5.5         | <code>api_post_ok</code>     | 5    | 100   | 0     | 0        |
| GPT-5.5         | <code>api_argorder</code>    | 75   | 100   | 0     | 0        |
| GPT-5.5         | <code>enc_amount</code>      | 55   | 0     | 100   | 100      |
| GPT-5.5         | <code>trap_store_wire</code> | 0    | 100   | 0     | 100      |
| DeepSeek-V4-Pro | <code>api_post_ok</code>     | 0    | 75    | 0     | 0        |
| DeepSeek-V4-Pro | <code>api_argorder</code>    | 2    | 68    | 0     | 5        |
| DeepSeek-V4-Pro | <code>enc_amount</code>      | 0    | 0     | 68    | 65       |
| DeepSeek-V4-Pro | <code>trap_store_wire</code> | 0    | 98    | 0     | 98       |
| Kimi-K2.6       | <code>api_post_ok</code>     | 0    | 100   | 0     | 0        |
| Kimi-K2.6       | <code>api_argorder</code>    | 45   | 100   | 0     | 0        |
| Kimi-K2.6       | <code>enc_amount</code>      | 0    | 0     | 95    | 92       |
| Kimi-K2.6       | <code>trap_store_wire</code> | 0    | 100   | 0     | 100      |

## B Extended experimental record

### Generality and controls: full protocols

Four follow-up experiments address three threat axes: source-pair generality, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

**A second incomparable pair, different domain.** We repeated the protocol with a second hand-built pair on a private `cache` module:  $V_1$ , the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False “newly inserted” flag, a `CacheError` type), versus  $V_2$ , a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at  $n = 40$  the pair is again verified incomparable ( $L_D = +0.774 / +0.496$ , both directions clear of zero) with a clean double crossover ( $V_1$  solves the contract and trap cells,  $V_2$  solves key-normalization, all none-baselines floor at 0%), and the dominance control again verifies one-directionally. Anti-monotonicity reproduces in direction at  $n = 40$  (the superset collapses the contract cell  $80\% \rightarrow 28\%$ ,  $\Delta_T = -0.52$ ); re-running the arms at  $n = 80$  shows the collapse is

stable ( $80\% \rightarrow 29\%$ ,  $\Delta_T = -0.51$ ) with Clopper–Pearson upper bound  $-0.296$ , which verifies *strictly negative* harm ( $\Delta_T < 0$  at  $\geq 90\%$ ) but does not clear the pre-registered margin  $\tau = 0.30$ : the binding constraint is the  $V_1$  ceiling itself ( $80\%$ , versus pair-1’s  $\sim 100\%$ ), which keeps the bound wide. We report pair-2 harm as a verified sign, not a verified magnitude. The  $n = 80$  run also strengthens the dominance control to a fully verified one-directional dominance ( $U_D = 0.000$  with a verified strict win). The relevance trap for this pair was a null, its prompt vocabulary leaned toward  $V_1$  ( $\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$ ), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

**A length-matched padding control.** Anti-monotonicity compares  $W_1$  against the strictly longer  $W_{1+}$ , and added input length alone is known to degrade performance [Du et al., 2025], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm  $W_{\text{pad}}$ : filler text in the same “team context (private)” register as  $W_2$ , placed in the same position  $W_2$  occupies in  $W_{1+}$  (padding first,  $W_1$  last), character-matched to  $W_{1+}$  within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On single-shot Haiku at  $n = 40$  on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`,  $W_1$  98%,  $W_{\text{pad}}$  100%,  $W_{1+}$  10%; on `api_post_ok`,  $W_1$  50%,  $W_{\text{pad}}$  65%,  $W_{1+}$  10%; and the same-run interference verification clears on `api_argorder` ( $\Delta_T = -0.88$ , upper bound  $-0.63$ ). The anti-monotonicity harm is therefore attributable to  $W_2$ ’s signal content interacting with the fixed rule, not to added tokens. One residual axis is not isolated by this control: the padding is off-topic for the `ledger` module while  $W_2$  is on-topic, so “topical but verifier-irrelevant” text remains a smaller, untested middle case.

**A routing-instruction ablation: the cure presupposes the diagnosis.** If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm  $W_{1+}^{\text{instr}}$  appends to  $W_{1+}$  a single routing sentence stating which convention serves which task family (“for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks”). On single-shot Haiku at  $n = 40$  the instruction recovers most, but not all, of the collapse: on `api_argorder`,  $2\% \rightarrow 80\%$  ( $W_1$  alone: 100%), and on `api_post_ok`,  $22\% \rightarrow 68\%$  ( $W_1$  alone: 50%); both recoveries over  $W_{1+}$  are large (non-overlapping exact 95% intervals), while the residual gap below  $W_1$  alone on the carrying cell ( $80\%$  vs.  $100\%$ , intervals  $[0.62, 0.92]$  vs.  $[0.90, 1.00]$ ) is directional but not separated at this  $n$ . We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper’s thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency verification measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally,  $\delta_M$  with the routing note is a *different, more Blackwell-coherent rule*; the verified falsification of Theorem 3 concerns the base rule and is unaffected, and even under the instructed rule, keeping the dominated signal is at best equal and directionally worse than dropping it, so the selection-level conclusion, keep the non-dominated set rather than forcing the superset, survives the ablation intact.

**A structured-context ablation: segmentation is not the bottleneck.** To separate “the model cannot segment the two conventions” from “the model cannot route them to task families,” the arm  $W_{1+}^{\text{xml}}$  wraps the same two conventions, in the same order as  $W_{1+}$ , in labeled XML sections (`<wire_encoding_invariant>`, `<api_call_contract>`) with no routing statement. On single-shot Haiku at  $n = 40$  per cell, structure alone mitigates weakly: `api_argorder` 42% (vs.  $W_{1+}$  2%,  $W_{1+}^{\text{instr}}$  80%,  $W_1$  100%) and `api_post_ok` 25% (vs. 12%, 68%, 57%). The structure-only mitigation is separated from the raw superset on the carrying cell (exact 95% intervals  $[0.27, 0.59]$  vs.  $[0.001, 0.13]$ ) but remains far below  $W_1$  alone; on `api_post_ok` it is not separated from the raw superset. Protocol note: the  $W_{1+}^{\text{xml}}$  arm was measured in a clean arm-only re-run (its arm in the first run was invalidated by a transport-failure window and scored no completions; the re-run had zero transport errors), so its comparison against same-day baseline arms is across runs rather than within one run. The resulting mitigation hierarchy, raw superset  $<$  structure  $<$  routing  $<$  dropping the dominated signal, locates the failure in convention-to-task routing.

## C Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle (the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below) is available at <https://github.com/abilous-ti/blackwell-llm-context>. The harness (`harness/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate  $\hat{\delta}_{\mathcal{D}}$ , the uniform incomparability verification (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference verification, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic CLI (`claude -p`), the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

**Machine-checked proofs.** The bundle also contains `lean/ContextSelection.lean`, a Lean 4 formalization of the two impossibility results (Remark 3). It takes no dependency on Mathlib, so it is checked by `lean ContextSelection.lean` against a bare toolchain with no build step or network access; the version is pinned in `lean/lean-toolchain` and a continuous-integration job re-checks it on a clean machine at every change, failing the build if any `sorry` appears or if either theorem stops reporting an empty axiom footprint.

**Protocol.** The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four verified cells are `api_post_ok`, `api_argorder` (`memo` arrives via a parameter, forcing the keyword-only fact), `enc_amount`, and `trap_store_wire`; the arms are  $\{\text{none}, W_1, W_2, W_{1+}\}$ , with  $W_{\text{pad}}$  (length-matched neutral filler  $++W_1$ ) added for the padding control. All six models are queried in a single-shot completion (one call, no tools, code extracted from the returned text), and PASS is measured over  $n = 40$  i.i.d. completions per cell at  $\eta = 0.10$ .

**Decoding, seeding, and retries.** All runs use the provider’s default decoding configuration: the harness sets no explicit temperature, top- $p$ , or seed (the Claude CLI and the Azure/OpenAI-compatible launchers are invoked without sampling overrides), so the  $n$  draws per cell are independent

samples from each provider’s default decoder, each a fresh single-turn call with a 200-second timeout. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Where a transport-outage window hit a whole arm or a single cell, we re-measured it cleanly in isolation with the retry harness and report only the clean measurement: the Opus  $W_{1+}$  arm; the Opus  $W_2$ /trap\_store\_wire cell, roughly 18 of whose 40 first-run draws were outage-scored (inferred from per-draw cost); and the Haiku  $W_1$ /enc\_amount cell, whose first run predated the retry wrapper. One cell could not be re-measured at revision time: GPT-5.5  $W_2$ /trap\_store\_wire returned HTTP 500 after retries on an unknown number of draws and the Azure quota was unavailable, so its reported 0/40 still contains outage-scored draws. It is an expected-zero cell (the arm lacks the decisive fact, and it is 0% on all five other models) and lies on no binding bound, but for that one cell the reported  $n$  is an upper bound on the number of real draws.

**Artifacts.** Per-model run logs and result JSONs are in the artifact repository (<https://github.com/abilous-ti/blackwell-llm-context>) under the pattern `blackwell_*.log.json`: the six single-shot  $n = 40$  model runs (`blackwell_haiku_ss_n40`, `blackwell_sonnet_ss_n40`, `blackwell_opus_ss_n40` with its arm-only re-run `blackwell_opus_ss_W1plus` and the clean single-cell re-measurements `blackwell_opus_ss_W2_trap_rerun` and `blackwell_haiku_ss_W1_enc_rerun`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`), and the follow-up controls `blackwell_pad_n40b` (length-matched padding, 400 calls), `blackwell_pair2_n80` (second source pair), `blackwell_instr_n40` (routing-instruction ablation), `blackwell_xml_n40` / `blackwell_xml_arm_only` (structured-context ablation), `reranker_trap_n30` (LLM listwise reranker probe; `harness/measure_reranker_trap.py`), and `dense_trap.json` (dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; `harness/measure_dense_trap.py`). The full per-cell record is in `docs/EXPERIMENT-BLACKWELL.md` and the theory hardening (formalization and proof sketches) in `docs/BLACKWELL.md`.

**Compute and cost.** Total measured Anthropic spend was approximately \$130: the single-shot Haiku, Sonnet, and Opus runs (Opus the costliest, plus its arm-only re-run) and the follow-up controls (padding, second pair, routing, structured-context). The non-Anthropic runs (GPT-5.5, DeepSeek, Kimi) used Azure quota; the Azure API returns no cost field, so this figure is the Anthropic spend, not the total across all six models. The dense/cross-encoder probe ran locally on open weights at zero API cost.

**Reference verification.** The bibliography was assembled in part with automated literature search and every entry was subsequently verified against arXiv/DBLP/publisher records; all entries resolve to real papers, and the classics (Blackwell 1953; Le Cam 1964/1986; Torgersen 1991; Ethayarajh et al. 2022) were checked against the primary sources.